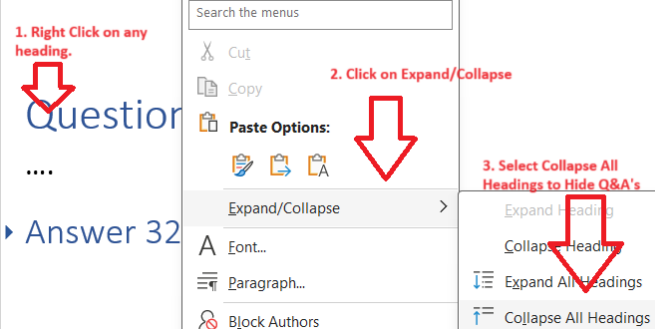


Instructions

Before taking the test, hide Questions and Answers by clicking on a heading → Expand Collapse and Select “Collapse All Headings.” Navigate through the Questions to read them and Expand the Answer to check if you are successful.	
MACRO ICONS: 	 : Insert a new blank question to fulfill.  : insert a “REVIEW” statement.
HashiCorp Terraform Associate 004 Exam HashiCorp Certified: Terraform Associate (004)  94% Students found the real exam almost same  1057 Students passed this exam after ExamTopic Prep  95.1% Average score during Real Exams at the Testing Centre Browse 128 Questions	

001 Question.

laC (Infrastructure as Code) can be stored in a version control system along with application code.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Infrastructure as Code (laC) is *meant* to be stored in version control systems (like Git) right alongside application code. This is actually one of its biggest strengths.

Discussion: <https://www.examttopics.com/discussions/hashicorp/view/385616-exam-terraform-associate-004-topic-1-question-1-discussion/>

002 Question.

It is best practice to store secret data in the same version control repository as your Terraform configuration.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: <https://www.examttopics.com/discussions/hashicorp/view/385617-exam-terraform-associate-004-topic-1-question-2-discussion/>

003 Question.

Which is an advantage of using IaC (Infrastructure as Code) that is not possible when provisioning with a GUI (Graphical User Interface)?

- A. Let's you version, reuse, and share infrastructure configuration.
- B. Secures your credentials.
- C. Provisions the same resources at a lower cost.
- D. Prevents manual modifications to your resources

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 1 question 3 discussion - ExamTopics](#)

004 Question.

What is an advantage of immutable infrastructure?

- A. Automatic infrastructure upgrades
- B. In-place infrastructure upgrades
- C. Quicker infrastructure upgrades
- D. Less complex infrastructure upgrades

Answer

Correct Answer: D

Explanation:

Immutable infrastructure means that instead of modifying existing servers, you replace them entirely with new, preconfigured ones. This approach removes a lot of the uncertainty and drift that comes from patching or updating systems in place.

Discussion: [Exam Terraform Associate 004 topic 1 question 4 discussion - ExamTopics](#)

005 Question.

What is the primary purpose of IaC (Infrastructure as Code)?

- A. To define a pipeline to test and deliver software.
- B. To provision infrastructure cheaply.
- C. To programmatically create and configure resources.
- D. To define a vendor-agnostic API.

Answer

Correct Answer: C

Explanation:

Discussion: <https://www.examttopics.com/discussions/hashicorp/view/385615-exam-terraform-associate-004-topic-1-question-5-discussion/>

006 Question.

Your team adopts an AWS CloudFormation as the standardized method for provisioning public cloud resources.

Which scenario presents a challenge for your team?

- A. Deploying new infrastructure into Microsoft Azure.
- B. Automating a manual, web console-based provisioning process.
- C. Building a reusable code base that can deploy resources into any AWS region.
- D. Managing a new application stack built on AWS-native services

Answer

Correct Answer: A

Explanation:

CloudFormation is an AWS-specific Infrastructure as Code service. It cannot deploy or manage resources in other cloud providers like Microsoft Azure or Google Cloud.

Discussion: <https://www.examttopics.com/discussions/hashicorp/view/385614-exam-terraform-associate-004-topic-1-question-6-discussion/>

007 Question.

Which is **not** a benefit of adopting IaC (Infrastructure as Code)?

- A. Reusability of code
- B. Automation
- C. A GUI (Graphical User Interface)
- D. Versioning

Answer

Correct Answer: C

Explanation:

Discussion: <https://www.examttopics.com/discussions/hashicorp/view/385613-exam-terraform-associate-004-topic-1-question-7-discussion/>

008 Question.

How does the use of Infrastructure as Code (IaC) enhance the reliability of your infrastructure?
(Choose two.)

- A. Configuration drift is reduced with declarative definitions.
- B. Updates are deployed with zero downtime.
- C. Proposed changes can be reviewed before being applied.
- D. Incorrect configurations cannot be deployed.
- E. Infrastructure is automatically scaled to meet demand.

Answer

Correct Answer: A,C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 1 question 8 discussion - ExamTopics](#)

009 Question.

Your team often uses API calls to create and manage cloud infrastructure.

In what ways does Terraform differ from conventional infrastructure management approaches?

- A. Terraform replaces cloud provider APIs with its own protocols, enabling automated deployments.
- B. Terraform describes infrastructure with version-controlled, repeatable configurations that specify the desired state.

- C. Terraform is merely a wrapper for cloud provider APIs, so there is little to no difference in calling the API directly.
- D. Terraform enforces infrastructure through imperative scripts to ensure tasks are completed in the proper order.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 1 question 9 discussion - ExamTopics](#)

010 Question.

Which of these workflows is only enabled by the use of Infrastructure as Code?

- A. Automatic scaling of resources based on application load.
- B. Cost optimization of infrastructure deployment.
- C. Role-based access control of cloud resources.
- D. Reviewing the proposed changes for potential security issues.

Answer

Correct Answer: D

Explanation:

Only Infrastructure as Code (IaC) enables a workflow where teams can **review proposed infrastructure changes before they are applied**. This is possible because IaC tools (like Terraform, Pulumi, CloudFormation) generate a *plan* or *diff* that shows exactly what will change. Teams can then perform code review, security review, and compliance checks just like with application code.

Discussion: [Exam Terraform Associate 004 topic 1 question 10 discussion - ExamTopics](#)

011 Question.

What does Terraform use to deploy infrastructure for different cloud providers?

- A. Custom APIs developed by HashiCorp
- B. Vendors' CLI tools
- C. Vendors' UI
- D. Vendor-specific providers

Answer

Correct Answer: D

Explanation:

Terraform deploys infrastructure through **vendor-specific providers**, which are plugins that know how to interact with each cloud provider's APIs (AWS, Azure, GCP, etc.). Providers translate Terraform's declarative configuration into the appropriate API calls.

Discussion: [Exam Terraform Associate 004 topic 1 question 11 discussion - ExamTopics](#)

012 Question.

How can you enable verbose logging to troubleshoot?

- A. Set the log level command line flag.
- B. Set the TF_LOG environment variable.
- C. Set the log level in your terraform block.

Answer

Correct Answer: B

Explanation:

Terraform's defining feature is that it uses **declarative, version-controlled configuration files** to describe the *desired state* of infrastructure. Terraform then determines how to reach that state by planning and applying changes.

Discussion: [Exam Terraform Associate 004 topic 2 question 1 discussion - ExamTopics](#)

013 Question.

Which command lets you experiment with Terraform expressions?

- A. terraform console
- B. terraform env
- C. terraform validate
- D. terraform test

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 2 discussion - ExamTopics](#)

014 Question.

You can install Community/Partner plugins using terraform init.

- A. True

B. False

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 3 discussion - ExamTopics](#)

015 Question.

What functionality do providers offer in Terraform? **(Choose three.)**

- A. Group a collection of Terraform configuration files that map to a single state file.
- B. Provision resources for public cloud infrastructure services.
- C. Interact with cloud provider APIs.
- D. Enforce security and compliance policies.
- E. Provision resources for on-premises infrastructure services.

Answer

Correct Answer: B,C,E

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 4 discussion - ExamTopics](#)

016 Question.

Terraform can only manage resource dependencies if you set them explicitly with the ***depends_on*** argument.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 5 discussion - ExamTopics](#)

017 Question.

Which of the following is not a valid Terraform collection type?

- A. set
- B. list
- C. map

D. tree

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 6 discussion - ExamTopics](#)

018 Question.

A resource block is shown in the Exhibit space of this page.

```
resource "aws_vpc" "main" {  
  name = "test"  
}
```

What is the provider for this resource?

- A. main
- B. vpc
- C. aws
- D. test

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 7 discussion - ExamTopics](#)

019 Question.

You must use different Terraform commands depending on the cloud provider you use.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 8 discussion - ExamTopics](#)

020 Question.

Terraform providers are always installed from the internet.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 9 discussion - ExamTopics](#)

021 Question.

You need to determine from which paths Terraform is loading the providers referenced in your *.tf files.

How can you enable additional logging to see this information?

- A. Set the environment variable TF_VAR_log=TRACE.
- B. Set the environment variable TF_LOG=TRACE.
- C. Set the environment variable TF_LOG=PATH.
- D. Set verbose logging for each provider in your Terraform configuration.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 10 discussion - ExamTopics](#)

022 Question.

Terraform requires using a different provider for each cloud provider where you want to deploy resources.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 11 discussion - ExamTopics](#)

023 Question.

Terraform cannot use a newly-defined cloud backend until it has been initialized with ***terraform init***.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Terraform **must** run terraform init before it can use a newly-defined backend (or any backend changes). Initialization downloads the backend plugin, configures it, and prepares the working directory. Without initialization, Terraform has no way to interact with that backend.

Discussion: [Exam Terraform Associate 004 topic 2 question 12 discussion - ExamTopics](#)

024 Question.

You have a list of numbers that represents the number of free CPU cores on each virtual cluster:

numcpus = [18, 3, 7, 11, 2]

What Terraform built-in function would you use to select the largest number from the list?

- A. top(numcpus)
- B. max(numcpus)
- C. ceil(numcpus)
- D. high[numcpus]

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 13 discussion - ExamTopics](#)

025 Question.

Which of these is stored in the **.terraform** directory?

- A. Providers and modules
- B. Lock file
- C. State file
- D. Configuration files

Answer

Correct Answer: A

Explanation:

The hidden .terraform directory (created automatically by terraform init) is Terraform's local cache for providers and modules.

Discussion: [Exam Terraform Associate 004 topic 2 question 14 discussion - ExamTopics](#)

026 Question.

```
variable "tags" {  
  description = "Metadata tags for resources"  
  type       = _____  
}
```

You will use this variable as the value for the tags argument in several resources. The data format must be a set of key value pairs.

Which type argument would you use?

- A. type = map(string)
- B. type = any
- C. type = list(string)
- D. type = object({ tags = string})

Answer

Correct Answer: A

Explanation:

A set of key-value pairs in Terraform is represented as a **map**, and when both keys and values are strings—as is typical for tags

Discussion: [Exam Terraform Associate 004 topic 2 question 15 discussion - ExamTopics](#)

027 Question.

Which of the following is allowed as a Terraform variable name?

- A. name
- B. source
- C. version
- D. count

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 2 question 16 discussion - ExamTopics](#)

028 Question.

terraform init creates an example **main.tf** file in the current directory.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 1 discussion - ExamTopics](#)

029 Question.

Which command must you run before you run a plan or apply for the first time?

- A. terraform validate
- B. terraform workspace
- C. terraform import
- D. terraform init

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 2 discussion - ExamTopics](#)

030 Question.

As a developer, you want to ensure your plugins are up-to-date with the latest versions. Which Terraform command should you use?

- A. terraform apply -upgrade
- B. terraform providers -upgrade
- C. terraform init -upgrade
- D. terraform refresh -upgrade

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 3 discussion - ExamTopics](#)

031 Question.

What does terraform apply change after you approve the execution plan? **(Choose two.)**

- A. The state file
- B. The cloud infrastructure
- C. The execution plan
- D. The Terraform code
- E. The .terraform directory

Answer

Correct Answer: A,B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 4 discussion - ExamTopics](#)

032 Question.

You modified your local Terraform configuration and ran terraform plan to review the changes. Simultaneously, your teammate manually modified the infrastructure component you are working on.

Since you already ran terraform plan locally, the execution plan for terraform apply will be the same.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

terraform apply always refreshes the state before creating the final plan.

Running terraform plan does not lock in the execution plan. If your teammate manually changes the infrastructure after you run plan but before you run apply, Terraform will detect those changes during the next refresh phase.

Discussion: [Exam Terraform Associate 004 topic 3 question 5 discussion - ExamTopics](#)

033 Question.

Which is the correct workflow for deploying new infrastructure with Terraform?

- A.
 1. Write Terraform configuration.
 2. Run terraform apply to create infrastructure.
 3. Use terraform validate to confirm Terraform deployed resources correctly.

- B.
 1. Write Terraform configuration.
 2. Run terraform init to initialize the working directory or workspace.
 3. Run terraform apply.
- C.
 1. Write Terraform configuration.
 2. Run terraform plan to initialize the working directory or workspace.
 3. Run terraform apply to create the infrastructure.
- D.
 1. Write Terraform configuration.
 2. Run terraform show to view proposed changes.
 3. Run terraform apply to create new infrastructure.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 6 discussion - ExamTopics](#)

034 Question.

After creating a new Terraform configuration, your config passes **terraform validate** but gives an “Access Denied” error from the cloud provider when running terraform plan.

Why didn't validate catch this issue?

- A. The working directory was not initialized, so the cloud provider plugin wasn't available to use when running the terraform validate command.
- B. The remote backend wasn't configured, so terraform validate couldn't load the state and detect the missing credentials.
- C. terraform validate only checks if a configuration is syntactically correct and internally consistent, and does not communicate with providers.
- D. Variables are only applied and validated during a terraform plan, so validate assumed defaults and returned the success message.

Answer

Correct Answer: C

Explanation:

terraform validate is **offline/static validation**. It checks that your Terraform configuration is:

- **syntactically valid** (HCL is well-formed)
- **internally consistent** (references, types, required arguments where possible)

But it **does not contact cloud providers** and **does not attempt authentication**. So it cannot detect “Access Denied” / missing credentials / insufficient permissions until Terraform actually tries to interact with the provider during terraform plan (or apply).

Discussion: [Exam Terraform Associate 004 topic 3 question 7 discussion - ExamTopics](#)

035 Question.

Only the user that generated a terraform plan may apply it.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 8 discussion - ExamTopics](#)

036 Question.

terraform apply will fail if you have not run terraform plan first to update the plan output.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 9 discussion - ExamTopics](#)

037 Question.

What's the proper syntax for the **plan** command?

- A. terraform plan -generate-config-out=tfplan
- B. terraform plan -target=tfplan
- C. terraform plan -out=tfplan
- D. terraform plan -var-file=tfplan

Answer

Correct Answer: C

Explanation:

terraform plan -out=tfplan: This tells Terraform to generate a plan and write it to a file named **tfplan**.

- **Why the others are incorrect**

- A. **-generate-config-out** This flag is used for *config generation mode*, not for saving a plan.
- B. **-target** This narrows the plan to a specific resource; it does *not* save a plan file.
- C. **-var-file** This loads variable definitions; again, it does *not* save a plan file.

Discussion: [Exam Terraform Associate 004 topic 3 question 10 discussion - ExamTopics](#)

038 Question.

Which is **true** about terraform apply? (**Choose two.**)

- A. It only operates on infrastructure defined in the current working directory or workspace.
- B. You cannot target specific resources for the operation.
- C. Depending on provider specification, Terraform may need to destroy and recreate your infrastructure resources.
- D. You must pass the output of a terraform plan command to it.
- E. By default, it does not refresh your state file to reflect the current infrastructure configuration.

Answer

Correct Answer: A,C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 11 discussion - ExamTopics](#)

039 Question.

Which syntax check returns an error when you run **terraform validate**?

- A. None of these will return an error.
- B. There is a missing variable block.
- C. The state file does not match the current infrastructure.
- D. The code contains tabs for indentation instead of spaces.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 12 discussion - ExamTopics](#)

040 Question.

What is the purpose of the **.terraform** directory in a Terraform workspace?

- A. The directory contains plugins and modules that Terraform downloads during initialization, along with other important information.
- B. The directory contains the provide credentials and the **.tfvars** files to prevent them from being committed to version control by accident.
- C. The directory is where Terraform creates and maintains the state file to track the underlying resources it creates and manages.
- D. The directory is used to convert, and store Terraform configuration files into API calls to communicate with the targeted platform.

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 13 discussion - ExamTopics](#)

041 Question.

How can **terraform plan** aid in the development process?

- A. Validates your expectations against the execution plan without permanently modifying state.
- B. Reconciles Terraform's state against deploys resources and permanently modifies state using the current status of deployed resources.
- C. Initializes your working directory containing your Terraform configuration files.
- D. Formats your Terraform configuration files.

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 14 discussion - ExamTopics](#)

042 Question.

You have a saved execution plan containing desired changes for infrastructure managed by Terraform. After running the command **terraform apply my.tfplan**, you receive the error shown in the Exhibit space on this page.

Error: Saved plan is stale

The given plan file can no longer be applied because the state was changed by another operation after the plan was created.

How can you apply the desired changes? (Choose two.)

- A. Generate a new execution plan file with **terraform plan**, and apply the new plan.

- B. Run terraform apply without the saved execution plan.
- C. Refresh the current state data using the -refresh-only flag.
- D. Force the apply command by adding the flag -lock=false.
- E. Update the current plan file using the terraform state push command.

Answer

Correct Answer: A,B

Explanation:

A. Generate a new execution plan file with terraform plan, **and apply the new plan.** A saved plan becomes invalid if the state changes. Terraform requires a *fresh* plan that matches the current state before applying.

B. Run terraform apply **without the saved execution plan.** Running terraform apply without specifying a plan forces Terraform to:

- Refresh the state
- Recalculate the plan
- Prompt you to approve the changes

This bypasses the stale plan entirely.

Discussion: [Exam Terraform Associate 004 topic 3 question 15 discussion - ExamTopics](#)

043 Question.

Which of these can you do with **terraform plan**? (Choose two.)

- A. Schedule Terraform to run at a planned time in the future.
- B. View the execution plan and check if the changes match your expectations.
- C. Execute a plan in a different workspace.
- D. Save a generated execution plan to apply later.

Answer

Correct Answer: B,D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 16 discussion - ExamTopics](#)

044 Question.

Which task does **terraform init** not perform?

- A. Sources any remote modules and downloads them.
- B. Connects to the configured backend.

- C. Validates that values are set for all required input variables.
- D. Sources all providers used in the configuration and downloads them.

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 17 discussion - ExamTopics](#)

045 Question.

terraform init retrieves and caches the configuration for all remote modules.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 18 discussion - ExamTopics](#)

046 Question.

You just upgraded the version of a provider in an existing Terraform configuration. What will install the new provider?

- A. Run ***terraform plan***.
- B. Run ***terraform apply***.
- C. Run ***terraform init -upgrade***.
- D. Run ***terraform refresh***

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 19 discussion - ExamTopics](#)

047 Question.

terraform validate confirms the syntax of Terraform files.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

terraform validate checks the **syntax and internal consistency** of your Terraform configuration files. It verifies that the files are syntactically valid and that references (variables, resources, providers) are well-formed, but it does **not** check whether resources can actually be created or whether credentials are valid.

Discussion: [Exam Terraform Associate 004 topic 3 question 20 discussion - ExamTopics](#)

048 Question.

A Terraform configuration is returning errors when running terraform plan.

How would you enable debug logging to investigate further?

- A. Rerun terraform plan with the correct flag.
- B. Edit the client configuration file.
- C. Update the backend configuration.
- D. Set the correct environment variable.

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 21 discussion - ExamTopics](#)

049 Question.

Why would you use the **-replace** flag for terraform apply?

- A. You want to force Terraform to destroy a resource on the next apply.
- B. You want Terraform to ignore a resource on the next apply.
- C. You want Terraform to destroy all the infrastructure in your workspace.
- D. You want to force Terraform to destroy and recreate a resource on the next apply.

Answer

Correct Answer: D

Explanation:

terraform apply -replace=RESOURCE tells Terraform:

“On the next apply, destroy this resource and recreate it, even if nothing in the configuration has changed.”

This is useful when a resource is in a bad state, drifted, or needs to be cycled for maintenance.

Discussion: [Exam Terraform Associate 004 topic 3 question 22 discussion - ExamTopics](#)

050 Question.

How does Terraform determine dependencies between resources when it creates an execution plan?

- A. Terraform builds a resource graph based on your configuration and your state file (if present).
- B. Terraform requires resources in your configuration be listed in the order they will be created to determine dependencies.
- C. Terraform requires all dependencies between resources be specified using the **depends_on** parameter.
- D. Terraform requires resource dependencies be defined as modules and sourced in order.

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 3 question 23 discussion - ExamTopics](#)

051 Question.

Which of the following arguments are required when declaring a Terraform output?

- A. default
- B. value
- C. description
- D. sensitive

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 1 discussion - ExamTopics](#)

052 Question.

Outside of the **required_providers** block, Terraform configurations always refer to providers by their local names.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: <https://www.examttopics.com/discussions/hashicorp/view/385073-exam-terraform-associate-004-topic-4-question-2-discussion/>

053 Question.

The Exhibit section of this page shows part of a configuration you've been asked to update.

```
data "azurerm_resource_group" "example" {
  name = var.resource_group_name
}
resource "azurerm_virtual_network" "example" {
  name = _____
}
```

The name of the Azure Virtual Network should be set to the name of the resource group followed by a dash and the word "vnet".

Which expression fulfils this requirement?

- A. "\${azurerm_resource_group.example.name}-vnet"
- B. join("-",var.resource_group_name,"vnet")
- C. concat(data.azurerm_resource_group.example.name, "-", "vnet")
- D. "\${data.azurerm_resource_group.example.name}-vnet"

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 3 discussion - ExamTopics](#)

054 Question.

Your team is collaborating on infrastructure using Terraform and are formatting their code to follow Terraform language style conventions.

How can you update your code to meet these requirements?

- A. Run ***terraform validate*** prior to executing terraform plan or terraform apply.
- B. Replace all tabs with spaces within your Terraform configuration files.
- C. Terraform automatically formats configuration on terraform apply.
- D. Run ***terraform fmt*** to update your Terraform configurations.

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 4 discussion - ExamTopics](#)

055 Question.

A resource block is shown in the Exhibit space of this page.

```
resource "aws_instance" "web" {  
  count = 2  
  name = "terraform-${count.index}"  
}
```

How do you reference the name value of the second instance of this resource?

- A. aws_instance.web.*.name
- B. aws_instance.web[2].name
- C. element(aws_instance.web, 2)
- D. aws_instance.web[1].name
- E. aws_instance.web[1]

Answer

Correct Answer: D

Explanation:

Discussion: <https://www.examttopics.com/discussions/hashicorp/view/385580-exam-terraform-associate-004-topic-4-question-5-discussion/>

056 Question.

When using multiple configurations of the same Terraform provider, what meta-argument must you include in any non-default provider configurations?

- A. alias
- B. name
- C. depends_on
- D. id

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 6 discussion - ExamTopics](#)

057 Question.

You want to create a string that is a combination of a generated **random_id** and a **variable** and reuse that string several times in your configuration.

What is the simplest correct way to implement this without repeating the **random_id** and **variable**?

- A. Add an output value.
- B. Add a local value.
- C. Use a data source.
- D. Use a module.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 7 discussion - ExamTopics](#)

058 Question.

Your Terraform configuration manages a resource that requires maximum uptime. You need to update the resource, and when you run terraform plan, Terraform notes that the update requires the resource to be destroyed and recreated.

Which lifecycle rule can you add to the resource to reduce the downtime of the resource while still applying the update?

- A. prevent_destroy = true
- B. destroy = false
- C. create_before_destroy = true
- D. ignore_changes = all

Answer

Correct Answer: C

Explanation:

When Terraform determines that a change requires a **replacement** (destroy + recreate), adding this lifecycle rule tells Terraform to:

Create the new resource first, and only destroy the old one after the replacement is ready.

That typically minimizes (or can even eliminate) downtime compared to the default behaviour, which often destroys first and then creates.

Discussion: [Exam Terraform Associate 004 topic 4 question 8 discussion - ExamTopics](#)

059 Question.

The Terraform configuration shown in the Exhibit space on this page **will create a new AWS instance**.

```
data "aws_instance" "web" {  
  filter {  
    name = "tag:Name"  
    values = ["web"]  
  }  
}
```

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 9 discussion - ExamTopics](#)

060 Question.

What kind of configuration block will manage an infrastructure object with settings specified within the block?

- A. data
- B. locals
- C. provider
- D. resource

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 10 discussion - ExamTopics](#)

061 Question.

You can execute ***terraform fmt*** to standardize all Terraform configurations within the current working directory to Terraform's canonical format and style.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 11 discussion - ExamTopics](#)

062 Question.

A data source is shown in the Exhibit space of this page.

```
data "aws_ami" "web" {
  most_recent = true
  owners = ["self"]
  tags = {
    Name = "web-server"
  }
}
```

How do you reference the **id attribute** of this data source?

- A. data.aws_ami.web.id
- B. web.id
- C. aws_ami.web.id
- D. data.web.id

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 12 discussion - ExamTopics](#)

063 Question.

A resource block is shown in the Exhibit space of this page.

```
resource "kubernetes_namespace" "example" {
  name = "test"
}
```

How would you reference the attribute name of this resource in HCL?

- A. resource.kubernetes_namespace.example.name
- B. data.kubernetes_namespace.name
- C. kubernetes_namespace.example.name
- D. kubernetes_namespace.test.name

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 13 discussion - ExamTopics](#)

064 Question.

You need to deploy resources into two different regions in the same Terraform configuration using the block in the Exhibit space on this page.

```
provider "aws" {
  region = "us-east-1"
}
provider "aws" {
  region = "us-west-2"
}
```

What do you need to add to the provider configuration to deploy the resource to the us-west-2 AWS region?

A.	resource "aws_instance" "example-us-west-2" { ami = data.aws_ami.ubuntu.id instance_type = "t3.micro" }
B.	provider "aws" { region = "us-east-1" } provider "aws" "west" { region = "us-west-2" }
C.	provider "aws_west" { region = "us-west-2" }
D.	provider "aws" { region = "us-east-1" } provider "aws" { alias = "west" region = "us-west-2" }

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 14 discussion - ExamTopics](#)

065 Question.

Two resources blocks are shown in the Exhibit space on this page: **azurerm_linux_web_app**, and **azurerm_role_assignment**.

```
resource "azurerm_linux_web_app" "app" {
  name          = "example-app"
  resource_group_name = azurerm_resource_group.rg.name
  location      = azurerm_resource_group.rg.location
  service_plan_id = azurerm_service_plan.plan.id

  identity {
    type = "UserAssigned"
    identity_ids = [azurerm_user_assigned_identity.app.id]
  }
}

resource "azurerm_role_assignment" "kv_access" {
  scope          = azurerm_key_vault.kv.id
  role_definition_name = "Key Vault Secrets User"
  principal_id    = azurerm_user_assigned_identity.app.principal_id
}
```

When provisioned, the web app will use the role assignment during creation, so the role assignment must be created first.

How do you ensure the **azurerm_role_assignment resource is created first?**

- A. Add a **depends_on** argument to the **azurerm_linux_web_app**.
- B. Change the order of the **azurerm_linux_web_app** and **azurerm_role_assignment** blocks.
- C. Add a **create_before_destroy** argument to the **azurerm_role_assignment**.
- D. Add a count argument to both resources.

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 15 discussion - ExamTopics](#)

066 Question.

A Terraform local value can reference other Terraform local values.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 16 discussion - ExamTopics](#)

067 Question.

You want to define multiple data disks as nested blocks inside the resource block for a virtual machine.

What Terraform feature would help you define the blocks using the values in a variable?

- A. Count arguments
- B. Collection functions
- C. Local values
- D. Dynamic blocks

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 17 discussion - ExamTopics](#)

068 Question.

You are creating a reusable Terraform configuration and want to include an optional **billing_dept** tag so your Finance team can track team-specific spending on resources.

Which of the following billing_dept variable declarations will achieve this?

A.	<pre>variable "billing_dept" { default = "" }</pre>
B.	<pre>variable "billing_dept" { type = optional(string) }</pre>
C.	<pre>variable "billing_dept" { optional = true }</pre>

D.	<code>variable "billing_dept" { type = default }</code>
----	---

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 18 discussion - ExamTopics](#)

069 Question.

Part of a configuration is shown in the Exhibit space of this page.

```
data "vsphere_datacenter" "dc" {}  
resource "vsphere_folder" "parent" {  
  path = "Production"  
  type = "vm"  
  datacenter_id = _____  
}
```

You want to pass the **id** of the **vsphere_datacenter** data source to the **datacenter_id** argument of the **vsphere_folder** resource.

Which reference would you use?

- A. `data.vsphere_datacenter.dc`
- B. `data.vsphere_datacenter.dc.id`
- C. `vsphere_datacenter.dc.id`
- D. `data.dc.id`

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 19 discussion - ExamTopics](#)

070 Question.

Your Terraform configuration declares a variable, you want to enforce that its value meets your specific requirements, and you want to block the Terraform operation if it does not.

What should you add to your configuration?

- A. Add a top-level check block.
- B. Add a validation block to the variable block.
- C. Add a top-level validation block.

- D. Add a check block to the variable block.

Answer

Correct Answer: B

Explanation:

A **validation block inside a variable** is exactly how you enforce custom rules on variable values in Terraform. If the value doesn't meet your conditions, Terraform will **fail early** with your custom error message.

Discussion: [Exam Terraform Associate 004 topic 4 question 20 discussion - ExamTopics](#)

071 Question.

Running ***terraform fmt*** without any flags in a directory with Terraform configuration files will check the formatting of those files, but will never change them.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 21 discussion - ExamTopics](#)

072 Question.

You want to retrieve the ID of an existing cloud resource for use in your Terraform configuration.

Which block type would you use?

- A. resource
- B. import
- C. data
- D. output

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 4 question 22 discussion - ExamTopics](#)

073 Question.

Your configuration contains a module block that references a module from the Terraform registry, and sets the version argument to **1.0**. You just published a new version of the module and updated your configuration to point to version **1.1**.

Which command must be run to install the new version?

- A. terraform apply
- B. terraform init
- C. terraform plan
- D. terraform modules

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 1 discussion - ExamTopics](#)

074 Question.

If one of your modules uses a local value, you can expose that value to callers of the module by defining a terraform output in the module's configuration.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 2 discussion - ExamTopics](#)

075 Question.

If a module declares a variable without a default value, you must pass the value of the variable within the module block when you call the module in your configuration.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 3 discussion - ExamTopics](#)

076 Question.

When you run ***terraform apply***, the Terraform CLI will print output values from both the root module and any child modules.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 4 discussion - ExamTopics](#)

077 Question.

Which type of information does the Terraform Registry provide about the modules it hosts?

- A. Required input variables
- B. Outputs
- C. Optional input variables and default values
- D. All of these are provided

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 5 discussion - ExamTopics](#)

078 Question.

When you include a module block in your configuration that references a module from the Terraform Registry, the version attribute is required.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 6 discussion - ExamTopics](#)

079 Question.

A child module can always access variables declared in its parent module.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: ExamHeist

080 Question.

You are updating a child module with the resource block shown in the Exhibit space on this page.

```
resource "aws_instance" "example" {  
  ami = "ami-0a123456789abcdef"  
  instance_type = "t3.micro"  
}
```

The **public_ip** attribute of the resource needs to be accessible to the parent module.

How do you meet this requirement?

- A. Add a data source to the parent module.
- B. Create a local value in the child module.
- C. Add an import block to the parent module.
- D. Create an output in the child module.

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 8 discussion - ExamTopics](#)

081 Question.

Your configuration defines the module block shown in the Exhibit space on this page. The **web_stack** module accepts an input variable named **servers**.

```
module "web_stack" {  
  source = "../modules/web_stack"  
}
```

Which of the following changes to the module block sets the servers variable to the value of "3"?

A.	<code>module "web_stack" { source = "./modules/web_stack" servers = 3 }</code>
B.	<code>module "web_stack" { source = "./modules/web_stack" inputs.servers = 3 }</code>
C.	<code>module "web_stack" { source = "./modules/web_stack" inputs = { servers = 3 } }</code>
D.	<code>module "web_stack" { source = "./modules/web_stack" var.servers = 3 }</code>

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 9 discussion - ExamTopics](#)

082 Question.

Your team is using a version **3.1.4** of a module from the public Terraform Registry, and they're worried about possible breaking changes in future versions of the module.

Which version argument should you add to the module block to prevent newer versions from being used?

- A. version = "3.1.4"
- B. version = "< 3.2"
- C. version = "!=3.1.5"
- D. version = "~>3.1.4"

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 10 discussion - ExamTopics](#)

083 Question.

Which of the following locations can Terraform use as a private source for modules? **(Choose two.)**

- A. Public repository on GitHub.
- B. Internally hosted VCS (Version Control System) platform.
- C. Public Terraform Registry.
- D. Private repository on GitHub.

Answer

Correct Answer: B,D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 11 discussion - ExamTopics](#)

084 Question.

Your configuration defines the module block shown in the Exhibit space on this page.

```
module "production" {  
  source = "../modules/web_stack"  
}
```

This module declares an output named *hostnames*.

How do you access the value of this output?

- A. production.hostnames
- B. module.production.hostnames
- C. web_stack.hostnames
- D. module.web_stack.hostnames

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 5 question 12 discussion - ExamTopics](#)

085 Question.

Which argument can you set on a module block to prevent Terraform from updating the module's configuration during an init or get operation?

- A. count
- B. lifecycle
- C. source
- D. version

Answer

Correct Answer: C

Explanation:

Terraform lets you prevent module updates during terraform init or terraform get by **pinning the module source**. When the source argument points to an exact version, commit, or archive, Terraform will not update it unless you explicitly change the source value.

Example:

```
module "network" {  
  
    source = "git::https://github.com/myorg/network-module.git?ref=v1.2.3"  
  
}
```

Because the ref is fixed, Terraform will not fetch newer commits.

Why the other options don't work

- **A. count**: Controls how many instances of a module or resource to create, not updates.
- **B. lifecycle**: Only applies to resources, not modules.
- **D. version**: Only applies to registry modules, and even then Terraform may still check for updates unless the version is pinned.

Discussion: [Exam Terraform Associate 004 topic 5 question 13 discussion - ExamTopics](#)

086 Question.

You need to destroy all of the resources in your Terraform workspace, except for ***aws_instance.ubuntu[1]***, which you want to keep.

How can you tell Terraform to stop managing that specific resource without destroying it?

- A. Change the value of the count argument on the resource.
- B. Remove the resource block from your configuration.
- C. Run ***terraform state rm aws_instance.ubuntu[1]***.
- D. Use a moved block.

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 1 discussion - ExamTopics](#)

087 Question.

When you run ***terraform apply -refresh-only***, which of the following is not consulted by Terraform to update the state file?

- A. Authentication credentials for the provider.
- B. Terraform configuration files defining the resources.
- C. The actual infrastructure as reported by the provider API.
- D. The most recent state file.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 2 discussion - ExamTopics](#)

088 Question.

When declaring a variable, setting the **sensitive** argument to true will prevent the value from being stored in the state file.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 3 discussion - ExamTopics](#)

089 Question.

Which are benefits of migrating from a local state backend to a remote backend? **(Choose two.)**

- A. Eliminates the need to manage credentials when deploying infrastructure to multiple cloud providers.
- B. State locking that allows multiple team members to safely work on the same infrastructure.
- C. Faster plans and apply execution because the state is cached locally on the cloud provider.
- D. Guarantees that configuration drift cannot occur for the managed infrastructure.
- E. The ability to enable server-side encryption at rest

Answer

Correct Answer: B,E

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 4 discussion - ExamTopics](#)

090 Question.

How can you configure a Terraform workspace to store its state remotely? **(Choose two.)**

- A. Add a **cloud block** inside the terraform block.
- B. Set the **TERRAFORM_BACKEND** environment variable.
- C. Add a **backend block** inside the terraform block.
- D. Set the **TERRAFORM_CLOUD** environment variable.

Answer

Correct Answer: A,C

Explanation:

- **Add a cloud block inside the terraform block:** This is used when storing state in Terraform Cloud or Terraform Enterprise.
- **Add a backend block inside the terraform block:** This is used for all other remote backends (S3, Azure Storage, GCS, Consul, etc.).

Discussion: [Exam Terraform Associate 004 topic 6 question 5 discussion - ExamTopics](#)

091 Question.

What does state locking accomplish?

- A. Sets a password on the state file.
- B. Blocks Terraform commands from modifying the state file.
- C. Encrypts any credentials stored within the state file.
- D. Prevents accidental deletion of the state file.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 6 discussion - ExamTopics](#)

092 Question.

Why is it considered important to treat your Terraform state file as sensitive?

- A. It contains personal information about the last user to update it.
- B. It can contain information such as resource passwords and keys.
- C. It can be manually edited to change the deployed resources.
- D. It stores all environment variables from the machine that created it.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 7 discussion - ExamTopics](#)

093 Question.

Terraform stores the value of an output in its state file, even if the sensitive argument is set to true.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 8 discussion - ExamTopics](#)

094 Question.

You provisioned virtual machines (VMs) on **Google Cloud Platform** using the **gcloud** command line tool.

What must be done to manage these VMs using Terraform instead? **(Choose two.)**

- A. Add an import block to the configuration.
- B. Run ***terraform apply -refresh-only***.
- C. Add a resource block for the existing VM.
- D. Run ***terraform state pull***.

Answer

Correct Answer:

Explanation: A,C

Discussion: [Exam Terraform Associate 004 topic 6 question 9 discussion - ExamTopics](#)

095 Question.

Which is a recommended way to share Terraform state across teams?

- A. Use a remote state backend.
- B. Store ***terraform.tfstate*** state in **HashiCorp Vault**.
- C. No additional configuration is recommended.
- D. Store ***terraform.tfstate*** state in version control.

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 10 discussion - ExamTopics](#)

096 Question.

Once you configure a new Terraform backend with a terraform code block, which command should you use to migrate the state file?

- A. terraform init
- B. terraform state rm
- C. terraform push
- D. terraform apply

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 11 discussion - ExamTopics](#)

097 Question.

You want to bring in an existing database under Terraform management.

What information is required to create a new import block for the database? (Choose two.)

- A. The connection string that Terraform will use to connect and manage the database.
- B. The ID associated with the current database on the cloud provider.
- C. The database platform and version that the existing resource is running.
- D. The path to the .tf file that contains the database resource block.
- E. The destination resource address of the block that will manage the database.

Answer

Correct Answer: B,E

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 12 discussion - ExamTopics](#)

098 Question.

Terraform encrypts sensitive values stored in your state file.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 13 discussion - ExamTopics](#)

099 Question.

You've just finished refactoring part of your Terraform workspace's configuration to use a module to manage some of your resources. When you plan your changes, you notice that Terraform will destroy and recreate the affected resources. Doing so could cause unintended downtime in the application your workspace manages.

What supported approach should you take to complete the refactor without destroying and recreating your resources?

- A. Add moved blocks to your configuration to let Terraform know the new resource addresses for the affected resources.
- B. Open your cloud provider's console and rename the effected resources.
- C. Run the terraform console command to edit your workspace's state and update the resource names.
- D. Manually edit your **terraform.tfstate** file and update the resource names.

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 14 discussion - ExamTopics](#)

100 Question.

What is the purpose of **state locking** in a remote backend?

- A. Ensures only one instance of Terraform can modify state at a time.
- B. Creates a backup of the state file in a secure location.
- C. Encrypts the state in the remote backend.
- D. Requires every instance of Terraform uses the same provider version.

Answer

Correct Answer: A

Explanation:

State locking is a safety mechanism used by many remote backends (like S3 with DynamoDB, Terraform Cloud, Consul, etc.) to prevent concurrent operations from corrupting the Terraform state.

Discussion: [Exam Terraform Associate 004 topic 6 question 15 discussion - ExamTopics](#)

101 Question.

What Terraform command always causes a state file to be updated with changes that might have been made outside of Terraform?

- A. terraform plan -refresh-only
- B. terraform show -json
- C. terraform apply =lock=false
- D. terraform plan -target=state

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 16 discussion - ExamTopics](#)

102 Question.

When should you use the **force-unlock** command?

- A. When you see a status message stating that you cannot acquire the lock.
- B. When apply has failed due to a state lock.
- C. When automatic unlocking has failed.
- D. When you have a high priority change.

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 17 discussion - ExamTopics](#)

103 Question.

The **-refresh-only** parameter will update your state file when used with terraform plan.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 18 discussion - ExamTopics](#)

104 Question.

You corrected a typo in a resource name, changing it from **aws_s3_bucket.photoes** to **aws_s3_bucket.photos**. You want to update the Terraform state so that the existing resource is recognized under the new name, without destroying and recreating it.

Which configuration should you use?

- A. Remove the old resource from your configuration and re-import it.
- B. Run terraform apply -refresh-only.
- C. Do nothing - Terraform will automatically update the state.
- D. Add a moved block to your configuration.

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 19 discussion - ExamTopics](#)

105 Question.

What feature stops multiple users from operating on the Terraform state at the same time?

- A. Remote state storage
- B. Version control
- C. State locking
- D. Provider constraints

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 20 discussion - ExamTopics](#)

106 Question.

What does the import block do?

- A. Migrate resources from one Terraform state file to another.
- B. Install a module.
- C. Combine Terraform state files.
- D. Import existing resources into state.

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 21 discussion - ExamTopics](#)

107 Question.

You have successfully deployed an application that includes a resource with a public IP address. Due to a configuration oversight, no outputs were defined.

Using the Terraform CLI, how can you find the IP address of the resource that was deployed with minimum disruption to the application?

- A. Run ***terraform destroy*** then ***terraform apply*** and look for the IP address in ***stdout***.
- B. Run ***terraform state list*** to find the name of the resource, then ***terraform state show*** to find the attributes including public IP address.
- C. In a new folder, use the ***terraform_remote_state data source*** to load in the state file, then write an output for each resource that you find the state file.
- D. Run ***terraform output ip_address*** to view the result.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 22 discussion - ExamTopics](#)

108 Question.

It is _____ to change the Terraform backend from the default local backend to a different one after performing your first terraform apply.

- A. discouraged
- B. mandatory
- C. impossible
- D. optional

Answer

Correct Answer: A (or D)

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 23 discussion - ExamTopics](#)

109 Question.

Which is a benefit of the Terraform state file?

- A. A state file can schedule recurring infrastructure tasks.
- B. A state file is the desired state expressed by the Terraform code files.
- C. A state file is a source of truth for resources provisioned with Terraform.

D. A state file is a source of truth for resources provisioned with a public cloud console.

Answer

Correct Answer: C

Explanation:

Discussion: [Exam Terraform Associate 004 topic 6 question 24 discussion - ExamTopics](#)

110 Question.

terraform destroy is the only way to remove infrastructure with Terraform.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 7 question 1 discussion - ExamTopics](#)

111 Question.

A senior admin accidentally deleted some of your cloud instances.

What will Terraform do when you run `terraform apply`?

- A. Stop and generate an error message about the missing instances.
- B. Rebuild only the instances that were deleted.
- C. Build a completely brand-new set of infrastructure.
- D. Tear down the entire workspace's infrastructure and rebuild it.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 7 question 2 discussion - ExamTopics](#)

112 Question.

You have a Terraform configuration that defines a single virtual machine with no references to it. You've run `terraform apply` to create the resource, then removed the resource definition from your Terraform configuration file.

What will happen when you run `terraform apply` in the working directory again?

- A. Terraform will error.

- B. Nothing will happen.
- C. Terraform will remove the virtual machine from the state file, but the resource will still exist.
- D. Terraform will destroy the virtual machine.

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 7 question 3 discussion - ExamTopics](#)

113 Question.

If you manually destroy resources within your infrastructure, what is the best practice for reflecting this change in Terraform?

- A. Run terraform import to reflect the changes in Terraform state.
- B. Manually update the state file to remove the destroyed resources.
- C. Remove the resource definition from your file and run **terraform apply -refresh-only**.
- D. The change will happen automatically during the next terraform apply.

Answer

Correct Answer: C

Explanation:

When you manually destroy infrastructure outside of Terraform, the Terraform state becomes out of sync with reality. The best practice is to update your configuration so Terraform understands that the resource should no longer exist.

Discussion: [Exam Terraform Associate 004 topic 7 question 4 discussion - ExamTopics](#)

114 Question.

When a **check block's** assertion fails, Terraform blocks the current operation from executing.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Use the check block to validate your infrastructure outside of the typical resource lifecycle.

Terraform executes the check block as the last step of plan or apply operation, after Terraform has planned or provisioned your infrastructure. When a check block's assertion fails, Terraform reports a warning and continues executing the current operation.

Discussion: [Exam Terraform Associate 004 topic 7 question 5 discussion - ExamTopics](#)

115 Question.

You just scaled your VM infrastructure and realize you set the count variable to the wrong value. You correct the value and save your change.

What must you do next to make your infrastructure match your configuration?

- A. Inspect your Terraform state because you want to change it.
- B. Run terraform apply and confirm the planned changes.
- C. Inspect all Terraform outputs to make sure they are correct.
- D. Reinitialize because your configuration has changed.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 7 question 6 discussion - ExamTopics](#)

116 Question.

What does *terraform destroy* do?

- A. Destroys all Terraform code files in the current directory, leaves the state file intact.
- B. Destroys all infrastructure in the Terraform state file.
- C. Destroys the Terraform state file, leaves the infrastructure intact.
- D. Destroys all infrastructure in the configured Terraform provider.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 7 question 7 discussion - ExamTopics](#)

117 Question.

You have a simple Terraform configuration containing one VM (virtual machine) in a cloud provider. You run terraform apply and the VM is created successfully.

You delete the VM using the cloud provider console, then run terraform apply again without changing any Terraform code.

What happens next?

- A. Terraform will report an error.
- B. Terraform will recreate the VM.
- C. Terraform will remove the VM from the state file.

D. Terraform will not make any changes.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 7 question 8 discussion - ExamTopics](#)

118 Question.

You have never used Terraform before and would like to test it out using a shared team account for a cloud provider. The shared team account already contains **15 VMs** (virtual machines). You develop a new Terraform configuration containing **one VM**, perform terraform apply, and see that your VM was created successfully.

How do you delete the newly-created VM with Terraform?

- A. Delete the Terraform state file and execute terraform apply.
- B. The Terraform state file only contains one new VM. Execute terraform destroy.
- C. The Terraform state file contains all 16 VMs in the team account. Execute terraform destroy and select the newly-created VM.
- D. Delete the VM using the cloud provider console, and execute terraform apply to apply the changes to the Terraform state file.

Answer

Correct Answer: B

Explanation:

Discussion: [Exam Terraform Associate 004 topic 7 question 9 discussion - ExamTopics](#)

119 Question.

Your team uses HCP Terraform to manage your infrastructure. You need to make a change to an infrastructure stack running in a public cloud.

Which pattern follows Infrastructure as Code best practices for making a change?

- A. Make the change via the public cloud API endpoint.
- B. Use the public cloud console to make the change.
- C. Run the public cloud CLI tool to make the change.
- D. Submit a pull request and wait for an approved merge of the change.
- E. Clone the repository containing your infrastructure code and then run the code.

Answer

Correct Answer: D

Explanation:

Discussion: [Exam Terraform Associate 004 topic 8 question 1 discussion - ExamTopics](#)

120 Question.

Which of the following does **HCP Terraform** perform during a health assessment for a workspace?
(Choose two.)

- A. Sentinel policy checks
- B. Resource drift detection
- C. Estimate infrastructure cost
- D. Check block validation
- E. Terraform test execution

Answer

Correct Answer: A,B

Explanation:

HCP Terraform's health assessments evaluate the overall condition of a workspace, and two of the key checks it performs are:

- **Sentinel policy checks:** HCP Terraform runs Sentinel policies as part of its governance workflow. These policies can enforce rules such as naming standards, cost controls, or security requirements. Failing policies are surfaced in the health assessment.
- **Resource drift detection:** HCP Terraform can detect when real-world infrastructure has drifted from what Terraform expects. This is a core part of assessing workspace health.

Discussion: [Exam Terraform Associate 004 topic 8 question 2 discussion - ExamTopics](#)

121 Question.

You manage two workspaces in your HCP Terraform organization. The first workspace manages your network configuration. The second workspace manages your compute resources, and retrieves values from the networking workspace.

What HCP Terraform feature lets you run an apply operation on the compute workspace every time you update the networking workspace?

- A. Projects
- B. Policy
- C. Run tasks
- D. Run triggers

Answer

Correct Answer: D

Explanation:

Run triggers allow you to automatically start a run in one workspace (your compute workspace) whenever another workspace (your networking workspace) completes an apply. This is perfect when one workspace depends on outputs from another.

Discussion: [Exam Terraform Associate 004 topic 8 question 3 discussion - ExamTopics](#)

122 Question.

You can configure multiple cloud blocks in your Terraform configuration to connect your workspace to both **HCP Terraform** and your **Terraform Enterprise** instance.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

- Terraform allows only one cloud block in a configuration.
- You cannot connect a single workspace to both HCP Terraform and Terraform Enterprise at the same time.

Discussion: [Exam Terraform Associate 004 topic 8 question 4 discussion - ExamTopics](#)

123 Question.

Which HCP Terraform feature is recommended when you want to use the same cloud provider credentials across multiple workspaces?

- A. Variable sets
- B. Version Control System (VCS) integration
- C. The **`terraform_remote_state`** data source

Answer

Correct Answer: A

Explanation:

Discussion: [Exam Terraform Associate 004 topic 8 question 5 discussion - ExamTopics](#)

124 Question.

You are responsible for a set of infrastructure which is managed by two workspaces: **example-network** and **example-compute**.

The **example-compute** workspace uses data from output values configured in the **example-network** workspace, and must be deployed afterward.

Currently, this is a manual process:

- An operator deploys changes to the **example-network** workspace.
- They manually copy the output values from the **example-network** workspace to input variables configured for the **example-compute** workspace.
- They deploy the **example-compute** workspace.

Which HCP Terraform features can you use to automate this process? **(Choose two.)**

- A. A **health check** configured on the **example-compute** workspace to create a plan when HCP Terraform applies changes to the **example-network** workspace.
- B. A **run trigger** configured on the **example-network** workspace to automatically plan changes to the **example-compute** workspace after every apply.
- C. A **health check** configured on the **example-network** workspace to create a plan on the **example-compute** workspace when HCP Terraform applies changes to it.
- D. A **run trigger** configured on the **example-compute** workspace to automatically plan changes after HCP Terraform applies changes to the **example-network** workspace.
- E. A **tfe_outputs** data source configured in the **example-compute** workspace to automatically load output values from the **example-network** workspace.

Answer

Correct Answer: D,E

Explanation:

Discussion: [Exam Terraform Associate 004 topic 8 question 6 discussion - ExamTopics](#)

125 Question.

Where does HashiCorp recommend you store API tokens and other secrets within your team's Terraform workspaces? **(Choose three.)**

- A. In a **terraform.tfvars** file, checked into your version control system.
- B. In an **environment variable** and referenced with **TF_VAR_variablename**.
- C. In **HashiCorp Vault**.
- D. In a **plaintext** document on a shared drive.
- E. In an **HCP Terraform variable**, with the sensitive option checked.

Answer

Correct Answer: B,C,E

Explanation:

Discussion: [Exam Terraform Associate 004 topic 8 question 7 discussion - ExamTopics](#)

126 Question.

Which of these statements about **HCP Terraform workspaces** is false?

- A. You can trigger plans and applies using version control system integrations.
- B. HCP Terraform workspaces offer role-based access controls.
- C. You must use the CLI to switch between workspaces.
- D. HCP Terraform workspaces can securely store cloud credentials.

Answer

Correct Answer: C

Explanation:

This statement is **false** because HCP Terraform workspaces are **not** switched using the local CLI. Instead, each workspace in HCP Terraform is its own isolated environment in the cloud, and you select or trigger runs through:

- The HCP Terraform UI
- VCS integrations
- API calls
- CLI-driven runs that target a specific workspace

There is no concept of “switching” workspaces locally the way you do with terraform workspace in CLI-driven Terraform.

Discussion: [Exam Terraform Associate 004 topic 8 question 8 discussion - ExamTopics](#)

127 Question.

How does an HCP Terraform integration using the cloud block differ from remote backends, such as S3?

- A. Using the cloud block automatically encrypts state files at rest and in transit.
- B. Using the cloud block requires using Terraform Enterprise.
- C. The cloud block supports multiple backends simultaneously.
- D. Using the cloud block enables remote operations on infrastructure managed by HCP Terraform or Terraform Enterprise.

Answer

Correct Answer: D

Explanation:

A **cloud block** configures Terraform to use **HCP Terraform or Terraform Enterprise** as its backend. This not only stores state remotely but also enables **remote operations**, meaning:

- Plans and applies run in HCP Terraform/TFE

- No local state file
- No need for local credentials
- Full audit logs, RBAC, and policy enforcement

Remote backends like S3 only store state; they **do not** provide remote execution.

Discussion: [Exam Terraform Associate 004 topic 8 question 9 discussion - ExamTopics](#)

128 Question.

Which features do **HCP Terraform workspaces** provide that are not available in **Terraform Community Edition**? (Choose three.)

- A. Automatic detection of common security issues.
- B. Remote execution of Terraform operations.
- C. Support for multiple cloud providers.
- D. Store your configuration in a Version Control System (VCS).
- E. State versions and run history.
- F. Store Terraform and environment variables in variable sets.

Answer

Correct Answer: B,E,F

Explanation:

B. Remote execution of Terraform operations: HCP Terraform runs plans and applies remotely on its own infrastructure. Community Edition runs everything locally.

E. State versions and run history: HCP Terraform keeps a full history of runs, including logs, diffs, and state snapshots. Community Edition only keeps a single local state file unless you manually configure a backend.

F. Store Terraform and environment variables in variable sets: HCP Terraform lets you create variable sets that can be shared across multiple workspaces. Community Edition has no such feature.

Discussion: [Exam Terraform Associate 004 topic 8 question 10 discussion - ExamTopics](#)

129 Question.

Which statements are true? (Choose Two)

- A. A precondition in a resource block must pass for Terraform to create the resource.
- B. A precondition cannot reference attributes of another resource that are only known after apply
- C. When a resource has a count meta-argument, Terraform evaluates the precondition for each instance.

- D. If a resource's postcondition block fails, Terraform still creates the resources that depend on it.
- E. A resource can either have a precondition or a postcondition block, but not both.

Answer

Correct Answer: A,C

Explanation:

A: A precondition in a resource block must pass for Terraform to create the resource.

Terraform evaluates the precondition before creating the resource. If the precondition fails, the resource is not created.

C: When a resource has a count meta-argument, Terraform evaluates the precondition for each instance. If a resource uses count, Terraform evaluates the precondition separately for each instance.

Incorrect:

B: A precondition cannot reference attributes of another resource that are only known after apply. Incorrect. Preconditions can reference such attributes, but if the value is unknown during planning, Terraform defers evaluation until apply.

D: If a resource's postcondition block fails, Terraform still creates the resources that depend on it. If a postcondition fails, Terraform marks the resource as failed, and dependent resources are not created.

E: A resource can either have a precondition or a postcondition block, but not both. A resource can have both precondition and postcondition blocks defined.

Discussion: EXAMHEIST - 129

130 Question.

Your team often uses API calls to create and manage cloud Infrastructure. In what ways does Terraform differ from conventional Infrastructure management approaches?

- A. Terraform replaces cloud provider APIs with its own protocols, enabling automated deployments.
- B. Terraform describes infrastructure with version-controlled, repeatable configurations that specify the desired state.
- C. Terraform enforces infrastructure through imperative scripts to ensure tasks are completed in the proper order.
- D. Terraform is merely a wrapper for cloud provider APIs, so there is little to no difference in calling the API directly.

Answer

Correct Answer: B

Explanation:

Terraform uses declarative configuration files to define the desired state of infrastructure, enabling version control, consistency, and repeatability.

Discussion: EXAMHEIST - 130

131 Question.

Which of the following is not a way to trigger **terraform destroy** ?

- A. terraform destroy
- B. All of these will trigger terraform destroy
- C. terraform plan destroy
- D. terraform destroy-auto-approve.

Answer

Correct Answer: C

Explanation:

- A. **terraform destroy: Incorrect.** This command directly triggers destruction of managed infrastructure.
- B. **All of these will trigger terraform destroy: Incorrect.** Not all listed commands correctly trigger destroy.
- C. **terraform plan destroy: Correct.** This is not valid syntax. The correct flag would be terraform plan -destroy.
- D. **terraform destroy-auto-approve: Incorrect.** While written incorrectly, the valid command terraform destroy -auto-approve does trigger destroy.

Discussion: EXAMHEIST - 132

132 Question.

You are making changes to existing Terraform code to add some new infrastructure. When is the best time to run **terraform validate**?

- A. After you run terraform apply, so you can validate your Infrastructure.
- B. After you run terraform plan, so you can validate that your state file is consistent with your infrastructure.
- C. Before you run terraform apply, so you can validate your provider credentials.
- D. Before you run terraform plan, so you can validate your code syntax.

Answer

Correct Answer: D

Explanation:

Discussion: EXAMHEIST - 133

133 Question.

You've updated your Terraform configuration, and you need to preview the proposed changes to your infrastructure. Which command should you run?

- A. terraform validate
- B. terraform get
- C. terraform plan
- D. terraform show

Answer

Correct Answer: C

Explanation:

Discussion: EXAMHEIST - 141

134 Question.

Which are advantages of IaC (Infrastructure as Code) patterns instead of manual infrastructure management? **(Choose Two)**

- A. Eliminating the need for monitoring.
- B. Managing configuration with a GUI.
- C. Automating infrastructure deployments.
- D. Version controlling infrastructure changes.

Answer

Correct Answer: C,D

Explanation:

Discussion: EXAMHEIST - 145

135 Question.

You have set the **TF_LOG_PATH environment variable** for Terraform, and you would like to ensure the logs contain all debug-level messages and verbose process logs.

Which action should you take?

- A. Run the terraform output command.
- B. Add a log argument to the terraform block.
- C. Update the **Terraform CLI** configuration file.
- D. Set the **TF LOG** environment variable to **TRACE**.

Answer

Correct Answer: D

Explanation:

To capture all debug-level and verbose process logs, you set:

TF_LOG=TRACE

This ensures Terraform generates detailed logging at the TRACE level, which includes debug messages and internal operations. Combined with TF_LOG_PATH, the logs will be written to the specified file.

Discussion: EXAMHEIST - 147

136 Question.

You're refactoring your Terraform configuration and have moved resources from one large module to multiple smaller ones.

When running terraform plan, Terraform wants to destroy and recreate the resources.

What should you add to the configuration to preserve the existing resources?

- A. Add a **depends_on** argument to each resource you move.
- B. Add **moved blocks** for each resource.
- C. Add **lifecycle prevent_destroy = true** to each of the resources.

Answer

Correct Answer: B

Explanation:

Discussion: EXAMHEIST - 158

137 Question.

A module block is shown in the Exhibit space on this page.

```
module "vpc"{  
  source = "terraform-awsmodules/vpc/aws"  
  version = "~>4.0".  
}
```

```
} 
```

That module block limits the module version to major version 4.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: EXAMHEIST - 159

138 Question.

You want to split a large **main.tf** file into multiple files for better organization within the same working directory.

What impact should you expect when you run a terraform plan?

- A. Significant changes should be expected since moving resources between files can trigger a destroy/replacement.
- B. No impact since Terraform reads all **.tf** files as a single configuration when executing a terraform plan.
- C. Only minor impacts since Terraform parses the **main.tf** file first, which could change the order in which resources are created.
- D. The terraform plan will fail since variable and provider blocks must appear before resource blocks.

Answer

Correct Answer: B

Explanation:

Discussion: EXAMHEIST - 161

139 Question.

Your security team scanned some Terraform workspaces and found secrets stored in plaintext in state files.

How can you protect that data?

- A. Delete the state file every time you run Terraform.
- B. Store the state in an encrypted backend.
- C. Always store your secrets in a **secrets.tfvars** file.
- D. Edit your state file to scrub out the sensitive data.

Answer

Correct Answer: B

Explanation:

Discussion: EXAMHEIST - 163

140 Question.

You can reference a resource created with **for_each** using a splat (*) expression.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Splat doesn't work for 'for_each'

Discussion: EXAMHEIST - 164

141 Question.

You add a new resource to an existing Terraform configuration, but do not update the version constraint in the configuration. The existing and new resources use the same provider. The working directory contains a **.terraform.lock.hcl** file.

How will Terraform choose which version of the provider to use?

- A. Terraform will use the latest version of the provider for the new resource and the version recorded in the lock file to manage existing resources.
- B. Terraform will use the version recorded in your lock file.
- C. Terraform will check your state file to determine the provider version to use.
- D. Terraform will use the latest version of the provider available at the time you provision your new resource.

Answer

Correct Answer: B

Explanation:

The **.terraform.lock.hcl** file locks provider versions for your working directory. This ensures consistent behaviour across runs, so Terraform uses the same provider version for all resources, even when you add new resources. The version constraint in your configuration (required_providers) defines which versions are allowed but Terraform will prioritize the version

in the lock file if it satisfies the constraint. Terraform does not check the state file to pick a provider version—the state only tracks infrastructure resources. It does not automatically use the latest provider version unless you explicitly run `terraform init -upgrade` and the new version satisfies the constraints.

Discussion: EXAMHEIST – 165

142 Question.

All standard backend types support state storage, locking, and remote operations like plan, apply, and destroy.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Not all standard Terraform backends (i.e. local) support state locking or remote operations (plan, apply, destroy).

Discussion: EXAMHEIST – 166

143 Question.

Which argument(s) are **required** when declaring a Terraform variable?

- A. type
- B. default
- C. description
- D. All of the above
- E. None of the above

Answer

Correct Answer: E

Explanation:

You can optionally add type, default, or description, but they are not mandatory.

Discussion: EXAMHEIST – 168

144 Question.

terraform apply is failing with the following error.

Error loading state: AccessDenied: Access Denied status code: 403, request id: 288766CE5CCA24A0, host id: web.example.com

What next step should you take to determine the root cause of the problem?

- A. Run **terraform login** to reauthenticate with the provider.
- B. Review **/var/log/terraform.log** for error messages.
- C. Review **syslog** for Terraform error messages.
- D. Set **TF_LOG=DEBUG**.

Answer

Correct Answer: D

Explanation:

Run Terraform with debug logs to get more detailed output **TF_LOG=DEBUG terraform apply**

Discussion: EXAMHEIST – 169

145 Question.

Which Terraform command checks that your configuration syntax is correct?

- A. terraform show
- B. terraform init
- C. terraform fmt
- D. terraform validate

Answer

Correct Answer: D

Explanation:

Terraform validate → Check configuration files for syntax correctness and internal consistency

Discussion: EXAMHEIST – 170

146 Question.

How does the HCP Terraform/Terraform Cloud integration differ from backends such as S3, Consul, etc.?

- A. It can execute Terraform runs on dedicated infrastructure in HCP Terraform/Terraform Cloud.
- B. It doesn't show the output of a terraform apply locally.
- C. It is only available to paying customers.

D. All of the above.

Answer

Correct Answer: A

Explanation:

Discussion: EXAMHEIST – 171

147 Question.

Which of these actions will prevent two Terraform runs from changing the same state file at the same time?

- A. Delete the state before running Terraform.
- B. Refresh the state after running Terraform.
- C. Run Terraform with parallelism set to 1.
- D. Configure state locking for your state backend

Answer

Correct Answer: D

Explanation:

State locking ensures that: Only one Terraform process can modify the state at a time. It prevents race conditions and corruption of the state file. If a lock is in place, other runs will wait or fail until the lock is released

Discussion: EXAMHEIST – 172

148 Question.

You have used Terraform to create an ephemeral development environment in the cloud and are now ready to destroy all the infrastructure described by your Terraform configuration. To be safe, you would like to first see all the infrastructure that Terraform will delete.

Which command should you use to show all of the resources that will be deleted? **Choose two correct answers.**

- A. Run ***terraform state rm ****
- B. Run ***terraform show -destroy***
- C. Run ***terraform destroy***. This will first output all the resource that will be deleted before prompting for approval
- D. Run ***terraform plan -destroy***

Answer

Correct Answer: C,D

Explanation:

To see all resources that Terraform will delete, you can use either of these commands:

- `terraform plan -destroy` shows the plan of destruction and asks for confirmation before proceeding.
- `terraform destroy` also outputs the plan with confirmation before executing the deletion.

You should **not** use `terraform state rm` because it removes resources from the state file without destroying them.

Discussion: EXAMHEIST – 175

149 Question.

You created infrastructure outside the Terraform workflow that you now want to manage using Terraform. **Which command brings the infrastructure into Terraform state?**

- A. `terraform init`
- B. `terraform get`
- C. `terraform import`
- D. `terraform refresh`

Answer

Correct Answer: C

Explanation:

Discussion: EXAMHEIST – 176

150 Question.

A developer launched a VM (virtual machine) outside of the Terraform workflow and ended up with two servers with the same name. They don't know which VM Terraform manages but do have a list of all active VM IDs.

Which of the following methods could you use to discover which instance Terraform manages?

- A. Run **`terraform state rm`** on both VMs, then **`terraform apply`** to recreate the correct one that will be managed by Terraform.
- B. Run **`terraform state list`** to find the names of all VMs, then run **`terraform state show`** for each of them to find which VM ID that Terraform manages.

- C. Run a ***terraform apply -refresh*** to identify the virtual machine IDs that are already managed by Terraform.
- D. Modify the Terraform configuration to add an import block for both of the virtual machines.

Answer

Correct Answer: B

Explanation:

The terraform state list command lists all resources that are managed by Terraform in the current state file. The terraform state show command shows the attributes of a single resource in the state file. By using these two commands, you can compare the VM IDs in your list with the ones in the state file and identify which one is managed by Terraform.

Discussion: EXAMHEIST – 177

151 Question.

When do you need to explicitly execute Terraform in **refresh-only** mode?

- A. Before every ***terraform plan***.
- B. Before every ***terraform apply***.
- C. Before every ***terraform import***.
- D. None of the above.

Answer

Correct Answer: D

Explanation:

terraform refresh (or refresh-only mode in newer versions) is used to update the state file with the real-world infrastructure without changing resources.

Discussion: EXAMHEIST-178

152 Question.

Which method for sharing Terraform configurations fulfills the following criteria:

- Keeps the configurations confidential within your organization
- Supports Terraform's semantic version constraints
- Provides a browsable directory of your modules

- A. A Subfolder within a workspace
- B. A Git repository containing your modules

- C. HCP Terraform/Terraform Cloud private registry
- D. Public Terraform module registry

Answer

Correct Answer: C

Explanation:

This is the method for sharing Terraform configurations that fulfills the following criteria:

- Keeps the configurations confidential within your organization
- Supports Terraform's semantic version constraints
- Provides a browsable directory

Discussion: EXAMHEIST-179

153 Question.

Which of the following **is not** a way to trigger terraform destroy?

- A. Passing **--destroy** at the end of a plan request.
- B. Running **terraform destroy** from the correct directory and then typing **yes** when prompted in the CLI.
- C. Using the destroy command with **auto-approve**.

Answer

Correct Answer: A

Explanation:

Discussion: EXAMHEIST-180

154 Question.

What Terraform command always causes a state file to be updated with changes that might have been made outside of Terraform?

- A. terraform apply =lock=false
- B. terraform plan -target=state
- C. terraform plan -refresh-only
- D. terraform show -json

Answer

Correct Answer: C

Explanation:

terraform plan -refresh-only (or equivalently terraform apply -refresh-only) tells Terraform to refresh the state file by reconciling it with the actual infrastructure, without proposing or applying any changes.

This command always updates the state file with any changes that might have been made outside of Terraform (manual edits, console changes, etc.).

Discussion: EXAMHEIST-181

155 Question.

A developer on your team is going to tear down an existing deployment managed by Terraform and deploy a new one. However, there is a server resource named **aws_instance.ubuntu[1]** they would like to keep. What command should they use to tell Terraform to stop managing that specific resource?

- A. terraform plan rm aws_instance.ubuntu[1]
- B. terraform state rm aws_instance.ubuntu[1]
- C. terraform apply rm aws_instance.ubuntu[1]
- D. terraform destroy rm aws_instance.ubuntu[1]

Answer

Correct Answer: B

Explanation:

To tell Terraform to stop managing a specific resource without destroying it, you can use the terraform state rm command. This command will remove the resource from the Terraform state, which means that Terraform will no longer track or update the corresponding remote object. However, the object will still exist in the remote system, and you can later use terraform import to start managing it again in a different configuration or workspace. The syntax for this command is terraform state rm, where rm stands for remove, followed by the identifier of the resource instance to remove. For example:

terraform state rm aws_instance.ubuntu[1] will remove the second instance of aws_instance.ubuntu named ubuntu from the state.

Discussion: EXAMHEIST-182

156 Question.

You are writing Terraform configuration that calls a child module. This child module declares an output value.

How can you ensure that Terraform will print out this value when you run Terraform CLI commands such as terraform apply?

- A. Nothing. Terraform will automatically print out all output values from child modules.
- B. Nothing. Terraform cannot use output values from child modules.
- C. Declare a new output in your root configuration that references the module's output.
- D. None of the above.

Answer

Correct Answer: C

Explanation:

Terraform **does not automatically print child module outputs**.

Discussion: EXAMHEIST-184

157 Question.

You have provisioned some virtual machines (VMs) on Google Cloud Platform (GCP) using the **gcloud command line tool**. However, you are standardizing with Terraform and want to manage these VMs using Terraform instead.

What are the two things you must do to achieve this? **(Choose two)**

- A. Provision new VMs using Terraform with the same VM names.
- B. Use the terraform import command for the existing VMs.
- C. Write Terraform configuration for the existing VMs.
- D. Run the terraform import-gcp command.

Answer

Correct Answer: B,C

Explanation:

1. Write Terraform configuration that describes each existing VM you want to manage (resource type, name, required arguments). This configuration must exist in your root module before importing.
2. Run terraform import for each existing VM to map the real resource into the Terraform state under the resource address you declared in the config.

Discussion: EXAMHEIST-185

158 Question.

Where does the Terraform local backend store its state?

- A. In the **.terraform** directory
- B. In the **.terraform.lock.hcl** file
- C. In the **terraform.tfstate** file

D. In the **terraform.tfstate** directory

Answer

Correct Answer: C

Explanation:

This is where the Terraform local backend stores its state, by default, unless you specify a different file name or location in your configuration. The local backend is the simplest backend type that stores the state file on your local disk.

Discussion: EXAMHEIST-187

159 Question.

A module block is shown in the Exhibit space of this page.

```
module "consul" {  
  source = "hashicorp/consul/aws"  
}
```

When you use a module block to reference a module from the Terraform Registry such as the one in the example, how do you specify version 1.0.0 of the module?

- A. You cannot. Modules stored on the public Terraform Registry do not support versioning.
- B. Append **?ref=v1.0.0** argument to the source path.
- C. Add a **version = "1.0.0"** attribute to the module block.
- D. Nothing. Modules stored on the public Terraform module Registry always default to version 1.0.0.

Answer

Correct Answer: C

Explanation:

Modules published in the Terraform Registry **do support versioning** and Terraform lets you pin a specific version directly in the module block.

Example:
<pre>module "example" { source = "hashicorp/consul/aws" version = "1.0.0" }</pre>

- **Why the other options don't work**
 - **A** — Incorrect. Registry modules absolutely support versioning.

- **B** — Incorrect. **?ref=** is used for Git sources, not Registry modules.
- **D** — Incorrect. Registry modules do **not** default to version 1.0.0; they default to the latest version unless specified.

Discussion: EXAMHEIST-188

160 Question.

You cannot install third-party plugins using terraform init.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

You can install third-party plugins using terraform init, if you specify the plugin directory in your configuration or as a command-line argument. You can also use the terraform providers mirror command to create a local mirror of providers from any source.

Discussion: EXAMHEIST-189

161 Question.

You're writing a Terraform configuration that needs to read input from a local file called **id_rsa.pub**. Which built-in Terraform function can you use to import the file's contents as a string?

- A. `file("id_rsa.pub")`
- B. `templatefile("id_rsa.pub")`
- C. `fileset("id_rsa.pub")`
- D. `filebase64("id_rsa.pub")`

Answer

Correct Answer: A

Explanation:

- **file():** Reads the contents of a local file and returns it as a string. Perfect for importing the contents of `id_rsa.pub` (your public SSH key).
- **templatefile():** Used to render a template file with variables, not for simply reading raw contents.
- **fileset():** Returns a list of file names that match a pattern in a given directory, not file contents.

- **filebase64():** Reads a file and returns its contents as a Base64-encoded string, which is useful for binary files but not needed for plain text like a public key.

Discussion: EXAMHEIST-190

162 Question.

When you initialize Terraform, where does it cache modules from the public Terraform Registry?

- A. They are not cached.
- B. In the **/tmp** directory.
- C. In the **.terraform** sub-directory.
- D. In memory.

Answer

Correct Answer: C

Explanation:

Discussion: EXAMHEIST-191

163 Question.

Which Terraform collection type should you use to store key/value pairs?

- A. map
- B. tuple
- C. list
- D. set

Answer

Correct Answer: A

Explanation:

- **map:** A collection of key/value pairs. Keys are strings, and values can be any type. Perfect when you need to associate values with specific identifiers.
- **tuple:** An ordered sequence of elements of potentially different types, accessed by index.
- **list:** An ordered sequence of elements of the same type, accessed by index.
- **set:** An unordered collection of unique elements of the same type, without duplicates.

Discussion: EXAMHEIST-192

164 Question.

A Terraform backend determines how Terraform loads state and stores updates when you execute which command?

- A. apply
- B. destroy
- C. Both of these are correct.
- D. Neither of these are correct.

Answer

Correct Answer: C

Explanation:

A **Terraform backend** determines **how and where state is stored and loaded**. This matters whenever Terraform needs to **read or write state**, which happens during:

- **terraform apply** → Reads the current state to plan changes and then updates the state after applying.
- **terraform destroy** → Reads the state to know what resources exist and updates it after destroying them.

So, both commands interact with the backend to load and update state.

Discussion: EXAMHEIST-193

165 Question.

Which of these is true about Terraform's plugin-based architecture?

- A. You can create a provider for your API if none exists.
- B. Terraform can only source providers from the internet.
- C. All providers are part of the Terraform core binary.
- D. Every provider in a configuration has its own state file for its resources..

Answer

Correct Answer: A

Explanation:

This is a core benefit of Terraform's plugin-based architecture. Terraform providers are essentially plugins that extend Terraform's capabilities to interact with various APIs (cloud providers, SaaS services, custom internal systems). If a provider doesn't exist for a specific API you need to manage, you can absolutely write your own custom provider using Go. This is a powerful extensibility feature.

Discussion: EXAMHEIST-195

166 Question.

You are using a networking module in your Terraform configuration with the name `my_network`. In your main configuration, you are trying to access the `vnet_id` attribute from this module with the following code:

```
resource "aws_instance" "example" {
  ami = "ami-0c55b2a94c9b82a81"
  instance_type = "t2.micro"
  subnet_id = module.my_network.vnet_id
}
output "net_id" {
  value = module.my_network.vnet_id
}
```

When you run `terraform validate`, you encounter the following error:

```
Error: Reference to undeclared output value on main.tf line 12, in output "net_id":
12: value = module.my_network.vnet_id
```

What must you do to successfully retrieve the `vnet_id` value from your networking module?

- A. Define the attribute `vnet_id` as a variable in the networking module.
- B. Change the referenced value to `module.my_network.outputs.vnet_id`
- C. Define the attribute `vnet_id` as an output in the networking module.
- D. Change the referenced value to `my_network.outputs.vnet_id`

Answer

Correct Answer: C

Explanation:

To correctly access the `vnet_id` from the networking module, you need to define `vnet_id` as an output within the networking module's configuration. This makes the value available for use in other parts of your Terraform code, using the syntax `module.my_network.vnet_id`.

Discussion: EXAMHEIST-196

167 Question.

Where in your Terraform configuration do you specify a state backend?

- A. The data source block
- B. The terraform block
- C. The provider block
- D. The resource block

Answer

Correct Answer: B

Explanation:

Discussion: EXAMHEIST-197

168 Question.

Multiple team members are collaborating on infrastructure using Terraform and want to format their Terraform code following standard Terraform style convention.

How should they ensure the code satisfies conventions?

- A. Use ***terraform fmt***.
- B. Terraform automatically formats configuration on ***terraform apply***.
- C. Run ***terraform validate*** prior to executing ***terraform plan*** or ***terraform apply***.
- D. Replace all tabs with spaces.

Answer

Correct Answer: A

Explanation:

Discussion: EXAMHEIST-200

169 Question.

You decide to move a Terraform state file to Amazon S3 from another location. You write the code shown in the Exhibit space into a file called **backend.tf**.

```
terraform {  
  backend "s3" {  
    bucket = "my-tf-bucket"  
    region = "us-east-1"  
  }  
}
```

Which command will migrate your current state file to the new S3 backend?

- A. terraform refresh
- B. terraform init
- C. terraform push
- D. terraform state

Answer

Correct Answer: B

Explanation:

When you add or change a backend (e.g., moving state to Amazon S3 via backend.tf), **terraform init** initializes the working directory and the backend. If an existing local state file is present, terraform init will **detect the change** and **prompt to migrate** the current state to the new S3 backend. You can also run it non-interactively with:

terraform init -migrate-state

Discussion: EXAMHEIST-202

170 Question.

You are a member of an operations team that uses infrastructure as code (IaC) practices with HCP Terraform/Terraform Cloud. You are tasked with making a change to an infrastructure stack running in a public cloud.

Which pattern would follow IaC best practices for making a change?

- A. Make the change via the public cloud API endpoint
- B. Clone the repository containing your infrastructure code and then run the code
- C. Use the public cloud console to make the change after a database record has been approved
- D. Make the change programmatically via the public cloud CLI
- E. Submit a pull request and wait for an approved merge of the proposed changes.

Answer

Correct Answer: E

Explanation:

Changes are proposed in code, reviewed by peers, merged into the main branch, and then applied via CI/CD pipelines. This ensures traceability, auditability, and consistency.

Discussion: EXAMHEIST-207

171 Question.

terraform validate uses provider APIs to verify your infrastructure settings.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Discussion: EXAMHEIST-208

172 Question.

Terraform installs its providers during which phase?

- A. plan
- B. init
- C. refresh
- D. All of the above

Answer

Correct Answer: B

Explanation:

Discussion: EXAMHEIST-210

173 Question.

Which of the following is **not true** of Terraform providers?

- A. An individual person can write a Terraform Provider
- B. A community of users can maintain a provider
- C. HashiCorp maintains some providers
- D. Cloud providers and infrastructure vendors can write, maintain, or collaborate on Terraform providers
- E. None of the above

Answer

Correct Answer: E

Explanation:

All of the statements are true of Terraform providers. Terraform providers are plugins that enable Terraform to interact with various APIs and services. Anyone can write a Terraform provider, either as an individual or as part of a community. HashiCorp maintains some providers, such as the AWS, Azure, and Google Cloud providers. Cloud providers and infrastructure vendors can also write, maintain, or collaborate on Terraform providers, such as VMware, Oracle, and Alibaba Cloud providers.

Discussion: EXAMHEIST-211

174 Question.

HashiCorp Configuration Language (HCL) supports user-defined functions.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

HashiCorp Configuration Language (HCL) does not support user-defined functions. You can only use the built-in functions that are provided by the language. The built-in functions allow you to perform various operations and transformations on values within expressions. The general syntax for function calls is a function name followed by comma-separated arguments in parentheses, such as `sum(1, 2, 3)`. You can find the documentation for all of the available built-in functions in the Terraform Registry under the Packer Documentation, depending on which tool you are using.

Discussion: EXAMHEIST-212

Which of these statements about HCP Terraform/Terraform Cloud workspaces is **false**?

- A. They have role-based access controls
- B. You must use the CLI to switch between workspaces
- C. Plans and applies can be triggered via version control system integrations
- D. They can securely store cloud credentials

Correct Answer: B

Explanation:

The statement that you must use the CLI to switch between workspaces is **false**. Terraform Cloud workspaces can be managed via the Terraform Cloud UI, the Workspaces API, or the Terraform Enterprise Providers. CLI switching is optional and not mandatory.

Other statements are true:

- Terraform Cloud workspaces support **role-based access controls**.
- They allow **plans and applies triggered by VCS integrations**.
- They can **securely store sensitive variables and credentials encrypted at rest**.

Discussion: EXAMHEIST-213

175 Question.

Which of these are benefits of using Sentinel with HCP Terraform/Terraform Cloud? (**Choose three.**)

- A. You can check out and check in cloud access keys.
- B. You can restrict specific resource configurations, such as disallowing the use of **`CIDR=0.0.0.0/0`**.

- C. Sentinel Policies can be written in HashiCorp Configuration Language (HCL).
- D. You can enforce a list of approved AWS AMIs.
- E. Policy-as-code can enforce security best practices.

Answer

Correct Answer: B,D,E

Explanation:

Sentinel is a policy-as-code framework that allows you to define and enforce rules on your Terraform configurations, states, and plans. Some benefits of using Sentinel with Terraform Cloud/Terraform Enterprise are:

- You can restrict specific resource configurations, such as disallowing the use of CIDR 0.0.0.0/0, which would open your network to the entire internet.
- Policies as code can enforce security best practices, such as requiring encryption, authentication, or compliance standards.
- You can enforce a list of approved AWS AMIs, which are pre-configured images that contain the operating system and software you need to run your applications.

Discussion: EXAMHEIST-215

176 Question.

Which of the following is **available only** in **HCP Terraform workspaces** and **not in Terraform CLI**?

- A. Secure variable storage.
- B. Support for multiple cloud providers.
- C. Dry runs with terraform plan.
- D. Using one workspace's state as a data source for another.

Answer

Correct Answer: D

Explanation:

- A. **Secure variable storage** → Not unique to HCP Terraform. You can store sensitive variables locally with CLI (though less secure), and HCP Terraform adds encryption and team access controls, but secure variable handling exists in both.
- B. **Support for multiple cloud providers** → Available in both CLI and HCP Terraform. Terraform is provider-agnostic.
- C. **Dry runs with terraform plan** → Available in both CLI and HCP Terraform. terraform plan is a core CLI feature.
- D. **Using one workspace's state as a data source for another** → This is a remote state data source feature that is only available in HCP Terraform (and Terraform Cloud/Enterprise). It

allows one workspace to directly consume outputs from another workspace's state, enabling cross-workspace dependencies.

Discussion: EXAMHEIST-216

177 Question.

Which two steps are required to provision new infrastructure in the Terraform workflow? (**Choose two.**)

- A. import
- B. init
- C. apply
- D. plan
- E. validate

Answer

Correct Answer: B,C

Explanation:

Provisioning new infrastructure with Terraform follows a workflow, but only certain steps are strictly required to create resources:

B. terraform init

- Initializes the working directory.
- Downloads provider plugins and sets up the backend.
- Must be run before any other commands in a new or changed configuration.

C. terraform apply

- Actually provisions (creates/updates/destroys) infrastructure based on your configuration and state.
- This is the step that makes changes in the real world.

Discussion: EXAMHEIST-217

178 Question.

Which of the following should you add in the **required_providers** block to define a provider version constraint?

- A. version >= 3.1
- B. version = ">= 3.1"
- C. version ~> 3.1

Answer

Correct Answer: B

Explanation:

The `required_providers` block is used to specify the provider versions that the configuration can work with. The version argument accepts a version constraint string, which **must be enclosed in double quotes**. The version constraint string can use operators such as `>=`, `<=`, etc., to specify the minimum, maximum, or exact version of the provider. For example, `version = "~>3.1"` means that the configuration can work with any provider version that is 3.1 or higher.

Discussion: EXAMHEIST-218

179 Question.

What is the purpose of state in Terraform?

- A. State codifies the dependencies of related resources.
- B. State stores variables and lets you quickly reuse existing code.
- C. State lets you enforce resource configurations that relate to compliance policies.
- D. State maps real world resources to your configuration and keeps track of metadata.

Answer

Correct Answer: D

Explanation:

Discussion: EXAMHEIST-219

180 Question.

Your DevOps team is currently using the local backend for your Terraform configuration. You would like to move to a remote backend to store the state file in a central location. Which of the following backends **would not** work?

- A. Amazon S3
- B. HCP Terraform/Terraform Cloud
- C. Git

Answer

Correct Answer: C

Explanation:

This is not a valid backend for Terraform, as it does not support locking or versioning of state files. The other options are valid backends that can store state files in a central location.

Discussion: EXAMHEIST-221

181 Question.

Which parameters does **terraform import** require? (**Choose two.**)

- A. Provider
- B. Path
- C. Resource ID
- D. Resource address

Answer

Correct Answer: C,D

Explanation:

C and D are correct. **terraform import** <resource_address> <resource_id>

Discussion: EXAMHEIST-222

182 Question.

Where can Terraform not load a provider from?

- A. Plugins directory
- B. Provider plugin cache
- C. Official HashiCorp Distribution on releases.hashicorp.com
- D. Source code

Answer

Correct Answer: D

Explanation:

Terraform can load providers from:

- A. **Plugins directory** → **Yes**, Terraform looks in the local .terraform/plugins directory for provider binaries.
- B. **Provider plugin cache** → **Yes**, Terraform maintains a plugin cache to avoid redownloading providers.
- C. **Official HashiCorp distribution on releases.hashicorp.com** → **Yes**, Terraform automatically downloads providers from the official registry/distribution.
- D. **Source code** → **No**. Terraform cannot load a provider directly from source code. Providers must be compiled into binaries before Terraform can use them.

Discussion: EXAMHEIST-223

183 Question.

Which configuration consistency errors does **terraform validate** report?

- A. Terraform module isn't the latest version
- B. Differences between local and remote state
- C. Declaring a resource identifier more than once
- D. A mix of spaces and tabs in configuration files

Answer

Correct Answer: C

Explanation:

terraform validate reports configuration consistency errors such as declaring a resource identifier more than once. This means the same resource type and name combination is used for multiple resource blocks, which is invalid in Terraform.

It does **not** report errors related to module versions, state differences, or formatting issues like spaces and tabs.

Discussion: EXAMHEIST-225

184 Question.

In a HCP Terraform/Terraform Cloud workspace linked to a version control repository, speculative plan runs start automatically when you merge or commit changes to version control.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

- In HCP Terraform/Terraform Cloud, **speculative plan** runs (also called **plan-only** runs) **occur automatically when you open a pull request (PR) or propose changes** in your linked version control system (VCS), **not when you merge or commit changes**.
 - **Speculative plans** are designed to show what would happen if the proposed changes were applied, without actually applying them.
 - When you **merge changes**, Terraform Cloud typically triggers a **real plan and apply run** (depending on workspace settings), not a speculative plan.

Discussion: EXAMHEIST-228

185 Question.

Your risk management organization requires that new AWS S3 buckets must be private, and their contents must be encrypted at rest. How can HCP Terraform/Terraform Cloud automatically and proactively enforce this security control?

- A. Auditing cloud storage buckets with a vulnerability scanning tool
- B. By adding variables to each HCP Terraform/Terraform Cloud workspace to ensure these settings are always enabled
- C. With an S3 module with proper settings for buckets
- D. With a Sentinel policy, which runs before every apply

Answer

Correct Answer: D

Explanation:

The best way to automatically and proactively enforce the security control that new AWS S3 buckets must be private and encrypted at rest is with a **Sentinel policy**, which runs before every apply. Sentinel is a policy-as-code framework that allows you to define and enforce logic-based policies for your infrastructure. Terraform Cloud supports Sentinel policies for all operations and can run them before any Terraform plan or apply operation. You can write a Sentinel policy that checks the configuration of the S3 buckets and ensures that they have the proper settings for privacy and encryption and then assign the policy to your Terraform Cloud organization or workspace. This way, Terraform Cloud will prevent any changes that violate the policy from being applied.

Discussion: EXAMHEIST-230

186 Question.

If you don't use the local Terraform backend, where else can Terraform save resource state?

- A. In an environment variable.
- B. In memory.
- C. In a remote location configured in the **.terraformrc** file, such as HCP Terraform or a cloud storage system.
- D. In a remote location configured in the terraform block, such as HCP Terraform or a cloud storage system.

Answer

Correct Answer: D

Explanation:

- Terraform uses **backends** to store state. If you don't use the default local backend, you can configure a **remote backend** in the terraform block of your configuration.
- Examples of remote backends:
 - **HCP Terraform / Terraform Cloud**
 - **AWS S3**
 - **Azure Storage**
 - **Google Cloud Storage**
- This configuration is done in your .tf files, not in .terraformrc.

Discussion: EXAMHEIST-231

187 Question.

Which of the following statements about Terraform modules is **not true**?

- A. Modules can call other modules
- B. A module is a container for one or more resources
- C. Modules must be publicly accessible
- D. You can call the same module multiple times

Answer

Correct Answer: C

Explanation:

Modules **do not need to be publicly accessible**. They can be local or private, stored in version control systems, or shared within an organization.

Discussion: EXAMHEIST-232

188 Question.

Which statement describes a goal of infrastructure as code (IaC)?

- A. The programmatic configuration of resources.
- B. Defining a vendor-agnostic API.
- C. Write once, run anywhere.
- D. A pipeline process to test and deliver software.

Answer

Correct Answer: A

Explanation:

Infrastructure as Code (IaC) is all about defining and provisioning infrastructure **programmatically**, using code instead of manually configuring hardware or cloud resources. This enables automation, repeatability, version control, and consistency across environments.

Here's why the other options are incorrect:

- **Defining a vendor-agnostic API** → Not a core IaC goal. Some IaC tools are multi-cloud, but vendor-agnostic APIs are not the purpose of IaC.
- **Write once, run anywhere** → This applies more to application portability (e.g., Java or containers), not IaC.
- **A pipeline process to test and deliver software** → That describes CI/CD, not IaC (though IaC often integrates with pipelines).

Discussion: EXAMHEIST-235

189 Question.

Which backend does the Terraform CLI use by default?

- A. Depends on the cloud provider configured
- B. HTTP
- C. Remote
- D. HCP Terraform/Terraform Cloud
- E. Local

Answer

Correct Answer: E

Explanation:

This is the backend that the Terraform CLI uses by default, unless you specify a different backend in your configuration. The local backend stores the state file in a local file named `terraform.tfstate`, which can be used to track and manage the state of your infrastructure.

Discussion: EXAMHEIST-235

190 Question.

You must initialize your working directory before running ***terraform validate***.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

You must initialize your working directory before running `terraform validate`, as it will ensure that all the required plugins and modules are installed and configured properly. If you skip this step, you may encounter errors or inconsistencies when validating your configuration files.

Discussion: EXAMHEIST-237

191 Question.

When does Terraform create the **.terraform.lock.hcl** file?

- A. After your first **terraform plan**.
- B. After your first **terraform apply**.
- C. After your first **terraform init**.
- D. Whenever you enable state locking.

Answer

Correct Answer: C

Explanation:

Terraform automatically creates or updates the dependency lock file each time you run the terraform init command.

Discussion: EXAMHEIST-238

192 Question.

Which of these commands makes your code more human-readable?

- A. terraform validate
- B. terraform output
- C. terraform show
- D. terraform fmt

Answer

Correct Answer: D

Explanation:

The command that makes your code more human-readable is terraform fmt. This command is used to rewrite Terraform configuration files to a canonical format and style, following the Terraform language style conventions and other minor adjustments for readability. The command is optional, opinionated, and has no customization options, but it is recommended to ensure consistency of style across different Terraform codebases. Consistency can help your team understand the code more quickly and easily, making the use of Terraform linting important. You can run this command on your configuration files before committing them to source control or as part of your CI/CD pipeline.

Discussion: EXAMHEIST-241

193 Question.

If a DevOps team adopts **AWS CloudFormation** as their standardized method for provisioning public cloud resources, which of the following scenarios poses a challenge for this team?

- A. The team is asked to build a reusable code base that can deploy resources into any AWS region
- B. The team is asked to manage a new application stack built on AWS-native services
- C. The organization decides to expand into Azure and wishes to deploy new infrastructure using their existing codebase
- D. The DevOps team is tasked with automating a manual, web console-based provisioning process

Answer

Correct Answer: C

Explanation:

AWS CloudFormation is a service specifically designed for provisioning AWS resources. If the organization decides to expand into Azure and deploy new infrastructure using the existing codebase, the DevOps team will face a challenge, as CloudFormation does not support Azure or other cloud provider.

Discussion: EXAMHEIST-242

194 Question.

Which of these actions are forbidden when the Terraform state file is locked? (**Choose three.**)

- A. terraform destroy
- B. terraform fmt
- C. terraform state list
- D. terraform apply
- E. terraform plan
- F. terraform validate

Answer

Correct Answer: A,D,E

Explanation:

When the **Terraform state file is locked**, it means Terraform is preventing concurrent modifications to the state to avoid corruption. During this lock:

- **Forbidden actions (require state access):**
 - terraform destroy → Needs to read and update state to remove resources.
 - terraform apply → Needs to read and update state to apply changes.

- terraform plan → Refreshes and compares state with configuration, so it requires access to the locked state.
- **Allowed actions (do not require state access):**
 - terraform fmt → Only formats configuration files; does not touch state.
 - terraform state list → Reads state but does not require exclusive lock.
 - terraform validate → Validates configuration syntax; does not use state.

Discussion: EXAMHEIST-244

195 Question.

Which of the following commands would you use to access all of the attributes and details of a resource managed by Terraform?

- A. terraform state show "***provider_type.name***"
- B. terraform state list
- C. terraform get "***provider_type.name***"
- D. terraform state list "***provider_type.name***"

Answer

Correct Answer: A

Explanation:

This is the command that you would use to access all the attributes and details of a resource managed by Terraform, by providing the resource address as an argument. For example:

terraform state show aws_instance.example

Discussion: EXAMHEIST-245

196 Question.

Terraform providers are part of the Terraform core binary.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Terraform providers are **not** part of the Terraform core binary. Instead:

- **Terraform Core:** Handles the execution plan, state management, and dependency graph.

- **Providers:** Are separate plugins that Terraform uses to interact with specific APIs (e.g., AWS, Azure, GCP). They are downloaded dynamically when you run `terraform init`.

This modular approach allows Terraform to support many platforms without bloating the core binary.

Discussion: EXAMHEIST-246

197 Question.

Before you can use a new backend or HCP Terraform/Terraform Cloud integration, you must first execute ***terraform init***.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Discussion: EXAMHEIST-248

198 Question.

Which type of block fetches or computes information for use elsewhere in a Terraform configuration?

- A. local
- B. data
- C. resource
- D. provider

Answer

Correct Answer: B

Explanation:

Discussion: EXAMHEIST-251

199 Question.

How can a ticket-based system slow down infrastructure provisioning and limit the ability to scale?
(Choose two.)

- A. End-users have to request infrastructure changes
- B. Ticket-based systems generate a full audit trail of the request and fulfillment process

- C. Users can access a catalog of approved resources from a drop-down list in a request form
- D. The more resources your organization needs, the more tickets your infrastructure team has to process.

Answer

Correct Answer: A,D

Explanation:

- **A:** When end-users must submit tickets for infrastructure changes, it introduces manual steps and delays.
- **D:** As demand grows, the number of tickets increases, creating bottlenecks and limiting scalability.

Options **B** and **C** do not inherently slow down provisioning; they provide audit trails and resource catalogues, which can improve governance and usability.

Discussion: EXAMHEIST-252

200 Question.

Which provider authentication method prevents credentials from being stored in the state file?

- A. Using environment variables
- B. Specifying the login credentials in the provider block
- C. Setting credentials as Terraform variables
- D. None of the above

Answer

Correct Answer: A

Explanation:

Using environment variables for authentication prevents credentials from being stored in the state file, which is a best practice for security.

Discussion: EXAMHEIST-254

201 Question.

What is a Terraform provider **not responsible** for?

- A. Provisioning infrastructure in multiple cloud providers.
- B. Managing actions to take based on resource differences.
- C. Understanding API interactions with a hosted service.
- D. Managing resources and data sources based on an API.

Answer

Correct Answer: A

Explanation:

C. Understanding API interactions with a hosted service → Providers know how to talk to the APIs of the services they manage.

D. Managing resources and data sources based on an API → Providers define resources and data sources that map to API objects.

B. Managing actions to take based on resource differences → Providers implement CRUD (create, read, update, delete) logic so Terraform can reconcile differences between state and configuration

Discussion: EXAMHEIST-255

202 Question.

Anyone can publish modules to the public Terraform Registry.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

The Terraform Registry allows any user to publish and share modules. Published modules support versioning, automatically generate documentation, allow browsing version histories, show examples and READMEs, and more. Public modules are managed via Git and GitHub, and publishing a module takes only a few minutes. Once a module is published, releasing a new version of a module is as simple as pushing a properly formed Git tag.

Discussion: EXAMHEIST-257

203 Question.

What type of information can be found on the Terraform Registry when using published modules?

- A. Required input variables
- B. Optional input variables and default values
- C. Outputs
- D. All of the above
- E. None of the above

Answer

Correct Answer: D

Explanation:

The public Terraform Module Registry automatically exposes all the information about published modules, including required input variables, optional input variables and default values, and outputs. This helps users to understand how to use and configure the modules.

Discussion: EXAMHEIST-258

204 Question.

One cloud block always maps to a single HCP Terraform/Terraform Cloud workspace.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

In Terraform, a **cloud block** defines how Terraform connects to Terraform Cloud or HCP Terraform, but it is **not restricted to a single workspace**.

A single cloud block can:

- D. Be used across **multiple configurations**
- E. Point to **different workspaces** if the workspace name is overridden via CLI or environment variables
- F. Be reused in different modules or projects

Discussion: EXAMHEIST-259

205 Question.

Which of these are features of Terraform Cloud? (**Choose two**).

- A. Automated infrastructure deployment visualization.
- B. Automatic backups of configurations and state.
- C. A web-based user interface (UI).
- D. Remote state storage.

Answer

Correct Answer: C,D

Explanation:

These are features of Terraform Cloud, which is a hosted service that provides a web-based UI, remote state storage, remote operations, collaboration features, and more for managing your Terraform infrastructure.

Discussion: EXAMHEIST-261

206 Question.

Which of the following is **not** a key principle of infrastructure as code (IaC)?

- A. Self-describing infrastructure
- B. Idempotence
- C. Versioned infrastructure
- D. Golden images

Answer

Correct Answer: D

Explanation:

Infrastructure as Code (IaC) focuses on defining and managing infrastructure through code, enabling automation, consistency, and repeatability. The key principles include:

- **Self-describing infrastructure:** The code clearly defines the desired state of the infrastructure.
- **Idempotence:** Applying the same configuration multiple times results in the same state without unintended changes.
- **Versioned infrastructure:** Infrastructure definitions are stored in version control systems, allowing tracking and rollback.

Golden images, on the other hand, refer to pre-configured machine images used for deployments. While they can be part of a deployment strategy, they are not a core principle of IaC because IaC emphasizes dynamic configuration through code rather than relying on static images.

Discussion: EXAMHEIST-263

207 Question.

You use a Cloud provider account that is shared with other team members. You previously used Terraform to create a load balancer that listens on port 80. After some application changes, you updated the Terraform code to change the port to 443.

You run `terraform plan` and see that the execution plan shows the port changing from 80 to 443 like you intended, and step away to grab some coffee.

In the meantime, another team member manually changes the load balancer port to 443 through the Cloud provider console before you get back to your desk.

What will happen when you run terraform apply upon returning to your desk?

- A. Terraform will not make any changes to the load balancer and will update the state file to reflect the manual change.
- B. Terraform will fail with an error because the state file is no longer accurate.
- C. Terraform will change the load balancer port to 80 and then change it back to 443.
- D. Terraform will recreate the load balancer.

Answer

Correct Answer: A

Explanation:

A. Terraform will not make any changes to the load balancer and will update the state file to reflect the manual change.

Discussion: EXAMHEIST-264

208 Question.

When using Terraform to deploy resources into Azure, which scenarios are **true** regarding state files? **(Choose two)**

- A. Changing resources via the Azure Cloud Console records the change in the current state file.
- B. When you change a resource via the Azure Cloud Console. Terraform records the changes in a new state file.
- C. When you change a Terraform-managed resource via the Azure Cloud Console. Terraform updates the state file to reflect the change during the next plan or apply.
- D. Changing resources via the Azure Cloud Console does not update current state file.

Answer

Correct Answer: C,D

Explanation:

C. When you change a Terraform-managed resource via the Azure Cloud Console, Terraform updates the state file to reflect the change during the next plan or apply.

D. Changing resources via the Azure Cloud Console does not update current state file.

Discussion: EXAMHEIST-265

209 Question.

You have developed a new cloud-based service that uses proprietary APIs and want to use Terraform to create, manage, and delete users from the service. How can Terraform interact with the service?

- A. Terraform can manage users for any service that is hosted on a public cloud provider.
- B. Develop and publish a customer provider to interact with the service using its proprietary APIs.

Answer

Correct Answer: B

Explanation:

Terraform interacts with external systems via **providers**. If your service has proprietary APIs and no existing Terraform provider, you'll need to build a **custom Terraform provider** that implements CRUD operations for your resources (e.g., users) and translates Terraform configurations into API calls.

Discussion: EXAMHEIST-266

210 Question.

You have just developed a new Terraform configuration for two virtual machines with a cloud provider. You would like to create the infrastructure for the first time.

Which Terraform command should you run first?

- A. terraform apply
- B. terraform plan
- C. terraform show
- D. terraform init

Answer

Correct Answer: D

Explanation:

Discussion: EXAMHEIST-267

211 Question.

You want to define a single input variable to capture configuration values for a server. The values must represent memory as a number, and the server name as a string.

Which variable type could you use for this input?

- A. List
- B. Object
- C. Map
- D. Terraform does not support complex input variables of different types

Answer

Correct Answer: B

Explanation:

This is the variable type that you could use for this input, as it can store multiple attributes of different types within a single value. The other options are either invalid or incorrect for this use case.

Discussion: EXAMHEIST-268

212 Question.

terraform validate confirms that your infrastructure matches the Terraform state file.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

Terraform validate does **not** confirm that your infrastructure matches the Terraform state file. It only checks whether the configuration files in a directory are syntactically valid and internally consistent. To confirm that your infrastructure matches the Terraform state file, you need to use **terraform plan** or **terraform apply with the -refresh-only option**.

Discussion: EXAMHEIST-270

213 Question.

You can access state stored in a local backend by using the **terraform_remote_state** data source.

- A. True
- B. False

Answer

Correct Answer: B

Explanation:

The `terraform_remote_state` data source is used to access state stored in remote backends (such as S3, Azure Blob Storage, GCS, or Terraform Cloud/Enterprise).

It is not supported for the local backend. If your state is stored locally (`terraform.tfstate`), you cannot use `terraform_remote_state` to read it.

To access values from local state, you would typically reference outputs directly within the same configuration or workspace.

Discussion: EXAMHEIST-273

214 Question.

Your root module contains a variable named `num_servers`. Which is the correct way to pass its value to a child module with an input named `servers`?

- A. `servers = var(num_servers)`
- B. `servers = var.num_servers`
- C. `servers = num_servers`
- D. `${var.num_servers}`

Answer

Correct Answer: B

Explanation:

Discussion: EXAMHEIST-275

215 Question.

What task does the ***terraform import*** command perform?

- A. Imports resources from one Terraform state file to another.
- B. Imports existing resources into Terraform's state file.
- C. Imports a new Terraform module into Terraform's state file.
- D. Imports all infrastructure from the configured cloud provider.
- E. Imports provider configuration from one state file to another.

Answer

Correct Answer: B

Explanation:

terraform import is designed for one very specific job: **Take infrastructure that already exists in the real world and bring it under Terraform management by adding it to the state file**

Discussion: EXAMHEIST-276

216 Question.

What is one disadvantage of using dynamic blocks in Terraform?

- A. Dynamic blocks can construct repeatable nested blocks
- B. Terraform will run more slowly
- C. They cannot be used to loop through a list of values
- D. They make configuration harder to read and understand.

Answer

Correct Answer: D

Explanation:

This is one disadvantage of using dynamic blocks in Terraform, as they can introduce complexity and reduce readability of the configuration. The other options are either advantages or incorrect statements.

Discussion: EXAMHEIST-277

217 Question.

Terraform variables and outputs that set the description argument will store that description in the state file.

- A. True
- B. False

Answer

Correct Answer: A

Explanation:

Terraform **does** store variable and output descriptions in the state file. This is intentional—Terraform keeps metadata (including descriptions) so that tools and future runs can understand the meaning of values stored in state.

Discussion: EXAMHEIST-279

218 Question.

What does this code do?

```
terraform {  
  required_providers {  
    aws = "~> 3.0"  
  }  
}
```

- A. Requires any version of the AWS provider ≥ 3.0 and < 4.0
- B. Requires any version of the AWS provider ≥ 3.0
- C. Requires any version of the AWS provider ≥ 3.0 major release like 4.1
- D. Requires any version of the AWS provider ≥ 3.0

Answer

Correct Answer: A

Explanation:

The code specifies a version constraint that requires any version of the AWS provider greater than or equal to 3.0 but less than 4.0.

Discussion: EXAMHEIST-280

219 Question.

When developing a Terraform module, how would you specify the version when publishing it to the official Terraform Registry?

- A. Configure it in the module's Terraform code
- B. Mention it on the module's configuration page on the Terraform Registry
- C. The Terraform Registry does not support versioning modules
- D. Tag a release in your module's source control repository.

Answer

Correct Answer: D

Explanation:

This is how you specify a module's version when publishing it to the public Terraform Module Registry, as it uses the tags from your version control system (such as GitHub or GitLab) to identify module versions. You need to use semantic versioning for your tags, such as v1.0.0.

Discussion: EXAMHEIST-281

220 Question.

Where does the default local Terraform backend store?

- A. tfplan files
- B. State file
- C. Provider plugins
- D. Terraform binary

Answer

Correct Answer: B

Explanation:

The default "local" Terraform backend stores the state file in a local file named terraform.tfstate, which can be used to track and manage the state of your infrastructure.

Discussion: EXAMHEIST-284

221 Question.

What command can you run to generate DOT (Document Template) formatted data to visualize Terraform dependencies?

- A. terraform graph
- B. terraform refresh
- C. terraform output
- D. terraform show

Answer

Correct Answer: A

Explanation:

Discussion: EXAMHEIST-285

222 Question.

A resource block is shown in the Exhibit space of this page.

```
resource "azurerm_resource_group" "dev" {  
  name = "test"  
  location = "westus"  
}
```

What is the Terraform resource name of that resource block?

- A. azurerm
- B. azurerm_resource_group
- C. test
- D. dev

Answer

Correct Answer: D

Explanation:

Discussion: EXAMHEIST-286

223 Question.

Which syntax check errors when you run ***terraform validate***?

- A. The code contains tabs for indentation instead of spaces.
- B. There is a missing value for a variable.
- C. The state file does not match the current infrastructure.
- D. None of the above.

Answer

Correct Answer: D

Explanation:

terraform validate checks only the syntax and internal consistency of your Terraform configuration files.

It does not:

- Care about whether you use tabs or spaces (A). Terraform accepts both.
- Check whether variables have values (B). That's validated at plan/apply time, not during validate.
- Compare the state file with real infrastructure (C). That happens during terraform plan or terraform refresh.

Discussion: EXAMHEIST-292